

Expected Problems

for Science Quiz

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This document collects expected mathematics problems for Science Quiz. Problems are organized by type, and each solution immediately follows the corresponding problem. References to Mathematical Concepts refer to the corresponding part of *Mathematics for Science Quiz*.

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Standard Mathematical Problems

Problem 1. (1988 IMO, Vieta Jumping). Let a and b be positive integers s.t. $ab + 1$ divides $a^2 + b^2$. Show that

$$\frac{a^2 + b^2}{ab + 1}$$

is the square of an integer.

Solution. Let

$$k = \frac{a^2 + b^2}{ab + 1} \in \mathbb{N}.$$

Suppose, for contradiction, that k is not a perfect square. Among all positive integer solutions of

$$a^2 + b^2 = k(ab + 1),$$

choose one for which $a + b$ is minimal, and assume without loss of generality that $a \geq b$.

If $a = b$, then

$$k = \frac{2a^2}{a^2 + 1} < 2,$$

so $k = 1$, a contradiction. Hence $a > b$. Regard the equation as a quadratic equation in a :

$$x^2 - kbx + b^2 - k = 0.$$

One root is a , so the other root is

$$c = kb - a \in \mathbb{Z}.$$

We claim that $c > 0$. From

$$a(a - kb) = k - b^2,$$

if $a = kb$, then $k = b^2$, contrary to the assumption. If $a > kb$, then the left-hand side is at least $a \geq kb \geq k$, whereas the right-hand side is less than k , which is impossible. Therefore $a < kb$, and hence $c > 0$.

Since k is a positive integer that is not a perfect square, we have $k \geq 2$. Let

$$f(x) = x^2 - kbx + b^2 - k.$$

Since

$$f(b) = (2 - k)b^2 - k < 0$$

and the two roots are c and a with $a > b$, it follows that

$$0 < c < b.$$

By Viète's formulas, c is the other root, and therefore

$$c^2 + b^2 = k(cb + 1).$$

Thus (c, b) is another positive integer solution with

$$c + b < a + b,$$

contradicting the minimality of $a + b$. Hence k must be a perfect square. □

Problem 2. (Sophomore's Dream). Prove the identity

$$\int_0^1 x^{-x} dx = \sum_{n=1}^{\infty} \frac{1}{n^n}.$$

Solution. For $0 < x \leq 1$, we have $-\ln x \geq 0$, and hence

$$x^{-x} = e^{-x \ln x} = \sum_{n=0}^{\infty} \frac{x^n (-\ln x)^n}{n!}.$$

Every term is nonnegative, so Tonelli's theorem from Mathematical Concepts permits term-by-term integration:

$$\int_0^1 x^{-x} dx = \sum_{n=0}^{\infty} \frac{1}{n!} \int_0^1 x^n (-\ln x)^n dx.$$

In the n th integral, let $t = -(n+1) \ln x$. Then

$$\int_0^1 x^n (-\ln x)^n dx = \frac{1}{(n+1)^{n+1}} \int_0^{\infty} t^n e^{-t} dt = \frac{n!}{(n+1)^{n+1}}.$$

Therefore

$$\int_0^1 x^{-x} dx = \sum_{n=0}^{\infty} \frac{1}{(n+1)^{n+1}} = \boxed{\sum_{n=1}^{\infty} \frac{1}{n^n}}.$$

Numerically, the common value is approximately

$$1.291285997 \dots$$

□

Problem 3. (1982 IMO). A function

$$f: \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{\geq 0}$$

satisfies

$$f(m+n) - f(m) - f(n) \in \{0, 1\}$$

for all positive integers m and n . Moreover,

$$f(2) = 0, \quad f(3) > 0, \quad f(9999) = 3333.$$

Determine $f(1982)$.

Solution. Repeatedly applying the given condition shows that, for all positive integers r and t ,

$$rf(t) \leq f(rt) \leq rf(t) + r - 1.$$

Apply this estimate to

$$N = 9999 \cdot 1982$$

in two different ways. Using $t = 1982$ and then $t = 9999$ gives

$$9999f(1982) \leq f(N) \leq 9999f(1982) + 9998$$

and

$$1982 \cdot 3333 \leq f(N) \leq 1982 \cdot 3333 + 1981.$$

Consequently,

$$\frac{1982 \cdot 3333 - 9998}{9999} \leq f(1982) \leq \frac{1982 \cdot 3333 + 1981}{9999}.$$

The left endpoint lies strictly between 659 and 660, whereas the right endpoint lies strictly between 660 and 661. Since $f(1982)$ is an integer, it follows that

$$\boxed{f(1982) = 660}.$$

□

Problem 4. (Happy Ending Problem). Five points are given in the plane, no three of which are collinear. Prove that four of them are the vertices of a convex quadrilateral.

Solution. Using the convex-hull terminology from Mathematical Concepts, consider the convex hull of the five points. If the convex hull has at least four vertices, any four of its vertices, taken in cyclic order, form a convex quadrilateral.

It remains to consider the case in which the convex hull is a triangle with vertices A, B, C . Let the remaining two points be P and Q ; both lie in the interior of $\triangle ABC$. The line PQ contains none of A, B, C , so two of the three vertices lie in the same open half-plane determined by PQ . Relabel them as A and B .

The triangle $\triangle ABQ$ meets the line PQ only at Q , because A and B lie strictly on the same side of that line. Hence $P \notin \triangle ABQ$. Similarly, $Q \notin \triangle ABP$. Since A and B are vertices of the convex hull of all five points, neither lies in the triangle determined by the other three points. Consequently, all four points A, B, P, Q are vertices of their convex hull, which is a convex quadrilateral.

□

Problem 5. (Sylvester–Gallai Theorem). Let S be a finite set of points in the plane, not all collinear. Prove that there exists a line containing exactly two points of S .

Solution. Call a line determined by points of S *ordinary* if it contains exactly two points of S . Consider all pairs (P, ℓ) s.t. $P \in S$, the line ℓ is determined by at least two points of S , and $P \notin \ell$. Since S is finite and not collinear, the set of positive distances arising in this way is finite and nonempty. Choose (P, ℓ) so that

$$h = \text{dist}(P, \ell) > 0$$

is as small as possible, and let H be the foot of the perpendicular from P to ℓ .

Suppose, for contradiction, that ℓ contains at least three points of S . We can choose distinct points $A, B \in S \cap \ell$ so that either $A = H$, or A and B lie on the same side of H with A closer to H . In either case,

$$AB < PB.$$

Indeed, if $A = H$, then $AB = HB < PB$; otherwise, $AB = HB - HA < HB < PB$.

Computing the area of $\triangle PAB$ with bases AB and PB gives

$$\frac{1}{2}ABh = [PAB] = \frac{1}{2}PB \text{dist}(A, PB).$$

Hence

$$\text{dist}(A, PB) = \frac{AB}{PB}h < h.$$

However, the line PB is determined by two points of S , and $A \notin PB$. This contradicts the minimality of h .

Therefore, ℓ contains exactly two points of S , so ℓ is the required ordinary line.

□

Problem 6. (Moser's Circle Problem). Let n be a positive integer. Place n points on a circle and join every pair of points by a chord. Assume that no three chords meet at a common point in the interior of the circle. Determine, with proof, the maximum number of regions into which the chords divide the circle.

Solution. There are

$$\binom{n}{2}$$

chords. Draw them one at a time. If a new chord contains r existing interior intersection points, those points divide it into $r + 1$ segments, and each segment splits one existing region into two. Therefore the new chord creates $r + 1$ new regions.

It follows that the final number of regions is

$$1 + |\{\text{chords}\}| + |\{\text{interior intersections}\}|.$$

Every interior intersection is determined by four boundary points: among the six chords joining those four points, exactly the two diagonals of their cyclic quadrilateral intersect in the interior. Conversely, every choice of four boundary points produces one such intersection. The assumption that no three chords are concurrent ensures that all these intersections are distinct. Hence there are

$$\binom{n}{4}$$

interior intersections.

Thus the maximum number of regions is

$$1 + \binom{n}{2} + \binom{n}{4}.$$

In particular, the first values are

$$1, 2, 4, 8, 16, 31, \dots,$$

so the apparent pattern 2^{n-1} fails when $n = 6$.

□

Problem 7. (1976 IMO). Determine, with proof, the largest number that can be written as the product of positive integers whose sum is 1976. The number of factors is not prescribed.

Solution. A maximizing product contains no factor 1, since replacing a pair $1, a$ by $a + 1$ preserves the sum and increases the product. It also contains no factor $k \geq 5$, because replacing k by 3 and $k - 3$ preserves the sum and increases the product:

$$3(k - 3) > k.$$

A factor 4 may be replaced by 2, 2 without changing the product. Thus it suffices to use only 2 and 3.

Three factors equal to 2 should be replaced by two factors equal to 3, since

$$2 + 2 + 2 = 3 + 3 \quad \text{and} \quad 2^3 < 3^2.$$

Hence at most two factors equal to 2 occur. Since

$$1976 = 3 \cdot 658 + 2,$$

the only optimal remainder pattern is one factor equal to 2 and 658 factors equal to 3.

Therefore, the largest possible product is

$$\boxed{2 \cdot 3^{658}}.$$

□

Problem 8. (2021 Putnam). A grasshopper starts at the origin in the coordinate plane and makes a sequence of hops. Each hop has length 5, and after every hop the grasshopper is at a point whose coordinates are both integers. Determine the smallest number of hops needed to reach

$$(2021, 2021).$$

Solution. If a hop has displacement vector $\mathbf{d} = (p, q)$, then

$$p^2 + q^2 = 25.$$

Thus the possible absolute coordinate pairs are $(5, 0)$ and $(4, 3)$, in either order. In particular, a single hop can increase the sum of the two coordinates by at most

$$4 + 3 = 7.$$

Since

$$2021 + 2021 = 4042 = 7 \cdot 577 + 3,$$

at least 578 hops are necessary.

This bound is attained because

$$(2021, 2021) = 288(3, 4) + 288(4, 3) + (5, 0) + (0, 5).$$

The right-hand side is a sum of

$$288 + 288 + 1 + 1 = 578$$

legal hop vectors. Therefore, the minimum number of hops is

$$\boxed{578}.$$

□

Problem 9. (Buffon's Needle). Infinitely many parallel lines are drawn in the plane, with distance d between consecutive lines. A needle of length ℓ , where

$$0 < \ell \leq d,$$

is dropped at random. Assume that its midpoint is uniformly distributed modulo one strip and, independently, that its direction is uniformly distributed. Determine the probability that the needle intersects one of the parallel lines.

Solution. Let X be the distance from the midpoint of the needle to the nearest parallel line, and let Θ be the acute angle between the needle and the lines. Then

$$0 \leq X \leq \frac{d}{2}, \quad 0 \leq \Theta \leq \frac{\pi}{2},$$

and X and Θ are independent and uniform on these intervals.

For a fixed angle θ , the needle intersects a line exactly when its perpendicular half-length reaches the nearest line, that is, when

$$X \leq \frac{\ell}{2} \sin \theta.$$

Since $\ell \leq d$, the right-hand side never exceeds $d/2$. Therefore,

$$\begin{aligned} \mathbb{P}(\text{intersection}) &= \frac{2}{\pi} \int_0^{\pi/2} \mathbb{P}\left(X \leq \frac{\ell}{2} \sin \theta\right) d\theta \\ &= \frac{2}{\pi} \int_0^{\pi/2} \frac{\ell \sin \theta}{d} d\theta \\ &= \boxed{\frac{2\ell}{\pi d}}. \end{aligned}$$

□

Problem 10. (2011 IMO, Windmill Problem). Let S be a finite set of at least two points in the plane, and assume that no three points of S are collinear. A windmill is the following process. Start with a line ℓ through a point $P \in S$. Rotate ℓ clockwise about the pivot P until it first meets another point $Q \in S$. The point Q then becomes the new pivot, and the line continues to rotate clockwise. The process is repeated indefinitely.

Prove that one can choose the initial point P and the initial line ℓ so that every point of S is used as a pivot infinitely many times.

Solution. Orient the rotating line and call its two open sides the left and right sides. When the pivot changes from T to U , retain the orientation of the line. Immediately after the change, the old pivot T lies on the same side on which U lay immediately before the change. Hence the ordered numbers of points on the two sides are invariant throughout the process, except at the instants when the line contains two points of S .

First suppose that

$$|S| = 2m + 1.$$

For every $T \in S$, there is an oriented line through T with exactly m points of S on each side. To see this, rotate a line through T by an angle π . The number of points on its left changes by one at each encounter with another point of S , and its initial and final values sum to $2m$; therefore the value m occurs.

Choose such a balanced line through any point as the initial state. The side counts remain (m, m) . Fix $T \in S$ and choose a balanced line r through T . Among all lines parallel to r , the line r is the unique one that contains a point of S and has m points on each side: translating it in either direction changes the balance at the first point crossed. Consequently, when the windmill becomes parallel to r , it must coincide with r , so T is a pivot. Thus every point is visited during each half-turn of the windmill line.

Now suppose that

$$|S| = 2m.$$

For every $T \in S$, the same half-turn argument gives an oriented line through T with $m - 1$ points on its left and m points on its right. Start the windmill from one such line. Its ordered side counts remain $(m - 1, m)$. For any prescribed T , choose an oriented line r through T with these counts. It is the unique oriented translate in its direction with these counts, so the windmill must pass through T when it reaches that oriented direction. Hence every point is visited during each full turn.

Only finitely many line directions are determined by pairs of points of S . Therefore the positive angular increments between successive pivot changes have a positive minimum. Since

the process continues indefinitely, the total angle of rotation tends to infinity. The preceding half-turn or full-turn argument then applies repeatedly, and every point of S is used as a pivot infinitely many times. □

Problem 11. (Art Gallery Problem). A museum has the shape of a simple polygon with n vertices. A guard must stand at a vertex of the polygon and can see a point whenever the entire line segment joining the guard to that point lies in the polygon. Prove that the entire museum can always be guarded by at most

$$\left\lceil \frac{n}{3} \right\rceil$$

guards.

Solution. Triangulate the polygon using nonintersecting diagonals between its vertices. The graph formed by the polygon sides and the triangulation diagonals admits a proper coloring with three colors.

To prove this, use induction on n . The assertion is immediate for a triangle. Every triangulated polygon has an ear: a triangle in the triangulation with two sides on the boundary. Remove the ear vertex and color the remaining triangulated polygon by induction. The two neighbors of the removed vertex are adjacent and therefore have different colors, so the removed vertex can be assigned the third color. This completes the induction.

Every triangle of the triangulation has one vertex of each color. Of the three color classes, one contains at most

$$\left\lceil \frac{n}{3} \right\rceil$$

vertices. Place guards at all vertices in that smallest color class. Every triangle contains one of these guards, and a guard at a vertex of a triangle sees every point of that triangle because a triangle is convex. Since the triangles cover the polygon, these guards see the entire museum. □

Problem 12. (Bertrand's Ballot Problem). In an election, candidate A receives p votes and candidate B receives q votes, where

$$p > q.$$

The $p + q$ ballots are counted in a uniformly random order. Determine the probability that, after every ballot is counted, the cumulative number of votes for A is strictly greater than the cumulative number of votes for B .

Solution. Represent a vote for A by a horizontal step and a vote for B by a vertical step. A counting order is then a lattice path from $(0, 0)$ to (p, q) , and the required condition is that the path remain strictly below the diagonal $y = x$ after leaving the origin.

The first ballot must be for A . Remove this first horizontal step and translate the remaining path one unit to the left. We obtain a path from $(0, 0)$ to $(p - 1, q)$ that never enters the region $y > x$. By the reflection principle, the number of such paths is

$$\binom{p+q-1}{q} - \binom{p+q-1}{q-1}.$$

Since all

$$\binom{p+q}{q}$$

ballot orders are equally likely, the desired probability is

$$\frac{\binom{p+q-1}{q} - \binom{p+q-1}{q-1}}{\binom{p+q}{q}} = \frac{p-q}{p+q}.$$

Therefore the probability is

$$\boxed{\frac{p-q}{p+q}}.$$

□

Problem 13. (1964 IMO). Seventeen people correspond with one another. Every pair discusses exactly one of three topics in their correspondence. Prove that there are three people s.t. every pair among them discusses the same topic.

Solution. Choose one person P . Among the 16 correspondences involving P , at least

$$\left\lceil \frac{16}{3} \right\rceil = 6$$

concern the same topic; call it topic 1, and let the six other people be $P_1, \dots, P_6$.

If some pair P_i, P_j also discusses topic 1, then P, P_i, P_j form the required triple. Suppose instead that no correspondence among $P_1, \dots, P_6$ concerns topic 1. Their mutual correspondences therefore use only topics 2 and 3.

It remains to use the elementary fact $R(3, 3) \leq 6$. Indeed, choose one of six vertices. At least three of its five incident edges have the same color. If one of the three edges joining their other endpoints has that color, there is a monochromatic triangle; otherwise those three edges form a triangle of the other color. Thus among $P_1, \dots, P_6$ there is a triangle all of whose edges have topic 2 or all of whose edges have topic 3.

□

Problem 14. (Sperner's Lemma). A triangle ABC is triangulated into finitely many smaller triangles. Label every vertex by one of 1, 2, 3 so that A, B, C receive labels 1, 2, 3, respectively; vertices on AB receive only 1 or 2; vertices on BC receive only 2 or 3; and vertices on CA receive only 3 or 1. Interior vertices may receive any label.

Prove that the number of small triangles whose three vertex labels are 1, 2, 3 is odd. In particular, at least one such triangle exists.

Solution. Call an edge a 12-edge if its endpoints have labels 1 and 2. Along the subdivided side AB , the sequence of labels begins with 1 and ends with 2. Hence the number of changes between 1 and 2 on AB is odd, so AB contains an odd number of 12-edges. The other two sides contain none.

Count incidences between small triangles and their 12-edges modulo 2. Every interior 12-edge is incident with two small triangles and contributes 0 modulo 2, whereas every boundary 12-edge contributes 1. Thus the total incidence count is odd.

A small triangle with labels 1, 2, 3 has exactly one 12-edge. A small triangle using only labels 1 and 2 has either zero or two such edges, and every other small triangle has none. Therefore the parity of the incidence count is exactly the parity of the number of fully labeled triangles. That number is consequently odd.

□

Problem 15. (Morley's Trisector Theorem). In a triangle ABC , trisect each interior angle. Let P be the intersection of the two trisectors adjacent to BC , let Q be the intersection of the two trisectors adjacent to CA , and let R be the intersection of the two trisectors adjacent to AB . Prove that $\triangle PQR$ is equilateral.

Solution. Write

$$\angle A = 3\alpha, \quad \angle B = 3\beta, \quad \angle C = 3\gamma.$$

Then

$$\alpha + \beta + \gamma = \frac{\pi}{3}.$$

Let $\mathcal{R}$ be the circumradius of $\triangle ABC$.

In $\triangle BCP$, the angles at B and C are β and γ . The sine rule and the identity

$$\sin(3t) = 4 \sin t \sin\left(\frac{\pi}{3} + t\right) \sin\left(\frac{\pi}{3} - t\right)$$

give

$$BP = \frac{BC \sin \gamma}{\sin(\beta + \gamma)} = 8\mathcal{R} \sin \alpha \sin \gamma \sin\left(\frac{\pi}{3} + \alpha\right).$$

Similarly, using $AB = 2\mathcal{R} \sin(3\gamma)$ in $\triangle ABR$,

$$BR = 8\mathcal{R} \sin \alpha \sin \gamma \sin\left(\frac{\pi}{3} + \gamma\right).$$

Since $\angle PBR = \beta$, the law of cosines yields

$$\begin{aligned} PR^2 &= (8\mathcal{R} \sin \alpha \sin \gamma)^2 \left(\sin^2\left(\frac{\pi}{3} + \alpha\right) + \sin^2\left(\frac{\pi}{3} + \gamma\right) \right. \\ &\quad \left. - 2 \sin\left(\frac{\pi}{3} + \alpha\right) \sin\left(\frac{\pi}{3} + \gamma\right) \cos \beta \right). \end{aligned}$$

Because

$$\left(\frac{\pi}{3} + \alpha\right) + \left(\frac{\pi}{3} + \gamma\right) + \beta = \pi,$$

the expression in parentheses is $\sin^2 \beta$. Hence

$$PR = 8\mathcal{R} \sin \alpha \sin \beta \sin \gamma.$$

Cyclically applying the same calculation gives the identical value for PQ and QR . Therefore $\triangle PQR$ is equilateral. □

Problem 16. (Steiner Tree for a Square). Determine the minimum possible total length of a connected road network containing all four vertices of a square of side length a . Roads may meet at additional points in the interior of the square.

Solution. The standard structure theorem for Euclidean Steiner minimal trees says that a shortest network is a tree, every nonterminal branching point has degree 3, and its three incident edges meet at angles of 120° . The possible square topologies have zero, one, or two Steiner points. With zero Steiner points, the shortest network is a minimum spanning tree of length $3a$. With one Steiner point, one first joins three square vertices by their three-terminal Steiner tree and then attaches the remaining vertex by an edge of length at least a . The three-terminal tree for a right isosceles triangle of legs a has length

$$a\sqrt{2 + \sqrt{3}},$$

so this case is also longer than the construction below. Thus it remains to examine the full topology with two Steiner points, each joined to two adjacent vertices and to the other Steiner point.

Place the square at

$$(0, -a/2), \quad (0, a/2), \quad (a, -a/2), \quad (a, a/2).$$

The two Steiner points lie on the horizontal symmetry axis. Write them as

$$S_1 = (d, 0), \quad S_2 = (a - d, 0).$$

The angle between the two edges from S_1 to the left-hand vertices is 120° . Taking their dot product gives

$$\frac{d^2 - a^2/4}{d^2 + a^2/4} = -\frac{1}{2},$$

so

$$d = \frac{a}{2\sqrt{3}}.$$

Each of the four terminal edges has length

$$\sqrt{d^2 + \frac{a^2}{4}} = \frac{a}{\sqrt{3}},$$

while

$$S_1 S_2 = a - 2d = a - \frac{a}{\sqrt{3}}.$$

Thus the total length is

$$4 \frac{a}{\sqrt{3}} + a - \frac{a}{\sqrt{3}} = \boxed{a(1 + \sqrt{3})}.$$

Since

$$a(1 + \sqrt{3}) < a \left(1 + \sqrt{2 + \sqrt{3}} \right) < 3a,$$

the two-Steiner construction is the global minimum. □

Problem 17. (1975 IMO). Let A be the sum of the decimal digits of 4444^{4444} , let B be the sum of the decimal digits of A , and let C be the sum of the decimal digits of B . Determine C .

Solution. Since $4444 < 10^4$,

$$4444^{4444} < 10^{17776}.$$

Hence

$$A \leq 9 \cdot 17776 = 159984.$$

Thus A has at most six digits, so $B \leq 54$. Among the integers from 1 to 54, the largest possible digit sum is 13, attained at 49. Therefore

$$1 \leq C \leq 13.$$

Repeated digit sums preserve a number modulo 9. Since

$$4444 \equiv 7 \pmod{9}, \quad 7^3 \equiv 1 \pmod{9}, \quad 4444 \equiv 1 \pmod{3},$$

we have

$$C \equiv 4444^{4444} \equiv 7 \pmod{9}.$$

The only integer between 1 and 13 congruent to 7 modulo 9 is

$$\boxed{C = 7}.$$

□

Problem 18. (Golomb's Tromino Theorem). Remove an arbitrary unit square from a $2^n \times 2^n$ board. Prove that the remaining board can be tiled by L-shaped trominoes, each consisting of three unit squares.

Solution. We use induction on n . For $n = 1$, the board is 2×2 , and the three remaining squares form one L-tromino.

Suppose the statement holds for $n - 1$. Divide a $2^n \times 2^n$ board into four $2^{n-1} \times 2^{n-1}$ quadrants. The missing square lies in one quadrant. Place one L-tromino at the center so that it covers one central square in each of the other three quadrants.

Each quadrant is now a $2^{n-1} \times 2^{n-1}$ board with exactly one square missing: the original missing square in one quadrant and a central square in each of the other three. By the induction hypothesis, every quadrant can be tiled. Together with the central tromino, these tilings cover the whole deficient board. □

Problem 19. A rectangle is tiled by finitely many smaller rectangles whose sides are parallel to the sides of the large rectangle. Each small rectangle has at least one side of integer length. Prove that the large rectangle also has at least one side of integer length.

Solution. Place the large rectangle at

$$[0, a] \times [0, b].$$

For any measurable region E , define

$$I(E) = \iint_E \sin(2\pi x) \sin(2\pi y) \, dx \, dy.$$

This quantity is additive over a finite dissection.

Consider a tile

$$[u, u + w] \times [v, v + h].$$

If w is an integer, then

$$\int_u^{u+w} \sin(2\pi x) \, dx = 0;$$

if h is an integer, the corresponding integral in y is zero. Therefore every tile has invariant 0, and additivity gives

$$I([0, a] \times [0, b]) = 0.$$

On the other hand,

$$I([0, a] \times [0, b]) = \frac{(1 - \cos(2\pi a))(1 - \cos(2\pi b))}{4\pi^2}.$$

Both factors in the numerator are nonnegative. Their product is zero only when

$$\cos(2\pi a) = 1 \quad \text{or} \quad \cos(2\pi b) = 1,$$

which means that a or b is an integer. □

Problem 20. (Hall's Marriage Theorem). A standard deck of 52 cards is divided into 13 piles of four cards. Prove that one can choose exactly one card from each pile so that the chosen cards have the thirteen distinct ranks

$$A, 2, 3, \dots, 10, J, Q, K.$$

Solution. Construct a bipartite graph whose left vertices are the 13 piles and whose right vertices are the 13 ranks. Join a pile to every rank that occurs in it. We verify Hall's condition.

Take any k piles. Together they contain $4k$ cards. If the union of their neighboring ranks contained at most $k - 1$ ranks, those ranks could account for at most

$$4(k - 1)$$

cards in the entire deck, since each rank occurs in four suits. This is impossible. Thus every k piles collectively contain at least k distinct ranks.

Hall's Marriage Theorem therefore gives a matching from all piles to distinct ranks. Choosing the matched card from each pile produces the required selection. $\square$

Problem 21. (Zeckendorf's Theorem). Define

$$F_1 = 1, \quad F_2 = 2, \quad F_{n+2} = F_{n+1} + F_n.$$

Prove that every positive integer can be written uniquely as a sum of nonconsecutive terms of this Fibonacci sequence.

Solution. For existence, apply the greedy algorithm. Given $N > 0$, choose the largest $F_k \leq N$. Since $N < F_{k+1} = F_k + F_{k-1}$, the remainder satisfies

$$0 \leq N - F_k < F_{k-1}.$$

Thus the next Fibonacci number selected by the same procedure has index at most $k - 2$. Repeating this process terminates and produces a sum of nonconsecutive Fibonacci numbers.

For uniqueness, use the identity

$$F_{k-1} + F_{k-3} + F_{k-5} + \cdots < F_k.$$

In fact, the sum on the left is $F_k - 1$. Suppose two valid representations differ, and let F_k be the largest term appearing in exactly one of them. The representation containing F_k exceeds the sum of every possible collection of nonconsecutive smaller terms, whereas the other representation uses only such smaller terms. They therefore cannot have the same value, a contradiction. $\square$

Problem 22. (1986 Putnam). Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices over a field, and suppose that

$$\mathbf{A} + \mathbf{B} = \mathbf{AB}.$$

Prove that

$$\mathbf{AB} = \mathbf{BA}.$$

Solution. Rearranging the hypothesis gives

$$(\mathbf{A} - \mathbf{I}_n)(\mathbf{B} - \mathbf{I}_n) = \mathbf{AB} - \mathbf{A} - \mathbf{B} + \mathbf{I}_n = \mathbf{I}_n.$$

For square matrices, a one-sided inverse is a two-sided inverse. Hence

$$(\mathbf{B} - \mathbf{I}_n)(\mathbf{A} - \mathbf{I}_n) = \mathbf{I}_n.$$

Expanding this identity gives

$$\mathbf{BA} - \mathbf{A} - \mathbf{B} = \mathbf{0}.$$

Comparing with the original relation yields

$$\mathbf{BA} = \mathbf{A} + \mathbf{B} = \mathbf{AB}.$$

$\square$

Problem 23. (Erdős–Szekeres Monotone Subsequence Theorem). Let

$$a_1, a_2, \dots, a_{n^2+1}$$

be distinct real numbers. Prove that the sequence contains an increasing subsequence of length $n + 1$ or a decreasing subsequence of length $n + 1$.

Solution. For each index i , let p_i be the length of the longest increasing subsequence ending at a_i , and let q_i be the length of the longest decreasing subsequence ending at a_i .

Suppose that neither required subsequence exists. Then

$$1 \leq p_i \leq n, \quad 1 \leq q_i \leq n$$

for every i . There are only n^2 possible ordered pairs (p_i, q_i) , so two indices $i < j$ have

$$(p_i, q_i) = (p_j, q_j).$$

Since the terms are distinct, either $a_i < a_j$ or $a_i > a_j$. In the first case, an increasing subsequence ending at a_i can be extended by a_j , so $p_j > p_i$. In the second case, $q_j > q_i$. Both alternatives contradict equality of the ordered pairs. □

Problem 24. (2002 Putnam). Given any five distinct points on a sphere, prove that some four of them lie in a closed hemisphere.

Solution. Choose two of the points that are not antipodal, and draw a great circle through them. The two chosen points lie on the boundary of each of the two closed hemispheres determined by that great circle.

Each of the remaining three points belongs to at least one of the two hemispheres. By the pigeonhole principle, one hemisphere contains at least two of those three points. Together with the two boundary points, it contains at least four of the five points. □

Problem 25. (Prouhet–Thue–Morse Partition). For a positive integer m , partition

$$\{0, 1, \dots, 2^m - 1\}$$

into sets A and B s.t.

$$\sum_{a \in A} a^r = \sum_{b \in B} b^r$$

for every $r = 0, 1, \dots, m - 1$.

Solution. Put an integer j in A when the number $s_2(j)$ of 1's in its binary expansion is even, and put it in B when $s_2(j)$ is odd. Consider

$$P(x) = \sum_{j=0}^{2^m-1} (-1)^{s_2(j)} x^j = \prod_{k=0}^{m-1} (1 - x^{2^k}).$$

Every factor on the right vanishes at $x = 1$, so P has a zero of order m there.

Apply the operator

$$x \frac{d}{dx}$$

exactly $r < m$ times and then set $x = 1$. Since the zero has order m , the result is 0. On the other hand,

$$\left(x \frac{d}{dx} \right)^r P(x) \Big|_{x=1} = \sum_{j=0}^{2^m-1} (-1)^{s_2(j)} j^r.$$

Separating the even and odd binary parities gives the required equal power sums. □

Problem 26. (Cayley's Formula). For an integer $n \geq 2$, determine the number of trees with vertex set

$$\{1, 2, \dots, n\}.$$

Solution. Associate a Prüfer sequence with a labeled tree as follows. Repeatedly delete the leaf with the smallest label and record its unique neighbor. Stop when two vertices remain. This produces a sequence of length $n - 2$, every term of which lies in $\{1, \dots, n\}$.

Conversely, start with a sequence

$$(p_1, p_2, \dots, p_{n-2}).$$

At each step, choose the smallest remaining label that does not occur in the current sequence, join it to the first term of the sequence, and then delete both that label and the first sequence term. After all terms have been removed, join the two labels that remain. This reconstructs a unique tree and reverses the encoding process.

Thus labeled trees are in bijection with length- $(n - 2)$ sequences over an alphabet of n labels. Their number is therefore

$$\boxed{n^{n-2}}.$$

□

Problem 27. (Tournament King Theorem). In a tournament, every pair of vertices is joined by exactly one directed edge. Prove that there is a vertex A s.t. every other vertex B is either an out-neighbor of A or an out-neighbor of some out-neighbor of A .

Solution. Choose a vertex A of maximum outdegree. Let B be any vertex that defeats A . If no vertex defeated by A defeats B , then B defeats A and every out-neighbor of A . Consequently,

$$\deg^+(B) \geq \deg^+(A) + 1,$$

contradicting the maximality of the outdegree of A .

Therefore, for every B that defeats A , there is a vertex C s.t. A defeats C and C defeats B . Hence A is a king. □

Problem 28. (Gale–Shapley Stable Marriage Problem). Two sets X and Y each contain n people. Every person has a strict preference ordering of all members of the other set. Prove that there is a perfect matching with no blocking pair, and give an algorithm that finds one.

Solution. Use the deferred-acceptance algorithm. While some person $x \in X$ is unmatched and has not proposed to everyone, let x propose to the most preferred member of Y who has not yet rejected x . Each $y \in Y$ keeps the most preferred proposal received so far and rejects every other proposer. A tentative engagement may therefore be replaced by a more preferred one.

Each ordered pair (x, y) is used in at most one proposal, so the algorithm terminates after at most n^2 proposals. At termination the matching is perfect: if some x were unmatched, then x would have proposed to every member of Y , so every member of Y would be holding a proposal; since the two sets have equal size, no member of X could then be unmatched.

Suppose that (x, y) is a blocking pair in the final matching. Since x prefers y to the assigned partner, x must have proposed to y earlier. The person y either rejected x immediately or later

replaced x by someone preferred to x . Since tentative partners of y only improve, the final partner is preferred to x , contradicting that (x, y) blocks the matching. $\square$

Problem 29. (Buffon's Noodle). Parallel lines at distance D apart are drawn in the plane. A rectifiable plane curve of total length L is placed with uniformly random translation modulo the strips and uniformly random orientation. If N is the number of intersections of the curve with the parallel lines, determine $\mathbb{E}[N]$.

Solution. First consider a line segment of length ℓ making an acute angle θ with the parallel lines. Its perpendicular span is

$$\ell \sin \theta.$$

Averaging over a uniformly random offset modulo D , the expected number of parallel lines crossed by this segment is

$$\frac{\ell \sin \theta}{D}.$$

Averaging over a uniformly random direction gives

$$\frac{2}{\pi} \int_0^{\pi/2} \frac{\ell \sin \theta}{D} d\theta = \frac{2\ell}{\pi D}.$$

For a polygonal curve, add the intersection counts of its segments. Linearity of expectation gives the same formula with ℓ replaced by the total length. A general rectifiable curve follows by polygonal approximation, which is the corresponding special case of the Cauchy–Crofton formula. Therefore,

$$\boxed{\mathbb{E}[N] = \frac{2L}{\pi D}}.$$

$\square$

Problem 30. (1987 IMO). Prove that there is no function

$$f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$$

s.t.

$$f(f(n)) = n + 1987$$

for every $n \in \mathbb{N}_0$.

Solution. Applying f to the given identity in two different ways gives

$$f(n + 1987) = f(f(f(n))) = f(n) + 1987.$$

Thus the residue of $f(n)$ modulo 1987 depends only on the residue of n modulo 1987.

For each $r \in \{0, 1, \dots, 1986\}$, write

$$f(r) = 1987q_r + s_r, \quad 0 \leq s_r < 1987.$$

Then

$$r + 1987 = f(f(r)) = 1987q_r + f(s_r).$$

Reducing modulo 1987 shows that

$$s_{s_r} = r.$$

Hence $r \mapsto s_r$ is an involution on a set of 1987 elements.

This involution has no fixed point. Indeed, if $s_r = r$, then the same displayed identity gives

$$1987(q_r + q_r) = 1987,$$

so $2q_r = 1$, which is impossible. An involution without fixed points partitions its underlying set into pairs, but a set of odd cardinality cannot be partitioned into pairs. Therefore no such function exists. $\square$

Problem 31. (1994 IMO). Let

$$A = \{a_1, a_2, \dots, a_m\} \subseteq \{1, 2, \dots, n\}$$

be a set of distinct positive integers. Suppose that whenever $x, y \in A$ and $x + y \leq n$, one also has $x + y \in A$. Prove that

$$\frac{a_1 + a_2 + \dots + a_m}{m} \geq \frac{n+1}{2}.$$

Solution. Let d be the least element of A . Partition A according to residue classes modulo d . Consider one nonempty class, and let b be its least element. Since $d, b \in A$, the closure condition implies that

$$b, \quad b + d, \quad b + 2d, \quad \dots$$

all belong to A for as long as they do not exceed n . Therefore this residue class of A is a complete arithmetic progression

$$b, b + d, \dots, c,$$

where c is the largest integer not exceeding n that is congruent to b modulo d .

We have $b \geq d$ and $c > n - d$, so

$$b + c > n.$$

Since $b + c$ is an integer,

$$b + c \geq n + 1.$$

The average of the progression is therefore at least $(n+1)/2$. Every residue class has average at least this value, so their weighted average, which is the average of all elements of A , also satisfies

$$\boxed{\frac{a_1 + a_2 + \dots + a_m}{m} \geq \frac{n+1}{2}}.$$

$\square$

Problem 32. (1974 IMO). Three cards bear distinct positive integers

$$0 < p < q < r.$$

In each round the cards are shuffled and dealt, one to each of three players A, B, C . Each player receives as many counters as the number on the card dealt to that player, and all counters are retained. At least two rounds are played.

At the end, the players have respectively

$$20, \quad 10, \quad 9$$

counters, and in the final round player B received the card bearing r . Determine who received the card bearing q in the first round.

Solution. Suppose that t rounds were played. Adding the final totals gives

$$t(p + q + r) = 20 + 10 + 9 = 39.$$

Since $t \geq 2$ and player C received at least one counter per round, $t \leq 9$. Hence

$$t = 3, \quad p + q + r = 13.$$

Player A received 20 counters in three rounds, so $r \geq 7$. In the last round player B received r , so the two earlier cards dealt to B have total $10 - r$. Checking the possibilities with $p < q < r$ and $p + q + r = 13$ gives the unique choice

$$(p, q, r) = (1, 4, 8).$$

Indeed, for $r = 7, 9$, or 10 , the required sum $10 - r$ cannot be formed by two cards, while for $r = 8$ it forces both earlier cards of B to be $p = 1$.

In each of the first two rounds, players A and C therefore received q and r . To reach a total of 20, player A must have received

$$r, r, q,$$

over the three rounds. Consequently player C received

$$q, q, p,$$

and in particular received q in the first round. Thus the required player is

$$\boxed{C}.$$

□

Problem 33. (2004 Putnam). A basketball player's cumulative free-throw percentage was below 80% at one point in a season and above 80% at a later point. Prove that at some intermediate point the cumulative percentage was exactly 80%.

Solution. After n attempts, let m be the number of successful free throws and define

$$D = 5m - 4n.$$

The cumulative percentage is below, equal to, or above 80% exactly when D is negative, zero, or positive, respectively.

After a successful free throw, D increases by

$$5 - 4 = 1,$$

whereas after a missed free throw, D decreases by 4. Thus the only way for the integer D to pass from a negative value to a positive value is through 0: an upward step has size only 1. Therefore at some intermediate point

$$5m - 4n = 0,$$

and the cumulative percentage was exactly $4/5 = 80\%$.

□

Problem 34. (2005 Putnam). Find a nonzero polynomial $P(x, y)$ in two variables s.t.

$$P(\lfloor a \rfloor, \lfloor 2a \rfloor) = 0$$

for every real number a .

Solution. Write

$$a = \lfloor a \rfloor + \{a\}, \quad 0 \leq \{a\} < 1.$$

Then

$$\lfloor 2a \rfloor - 2\lfloor a \rfloor = \lfloor 2\{a\} \rfloor \in \{0, 1\}.$$

Hence one may take

$$\boxed{P(x, y) = (y - 2x)(y - 2x - 1)}.$$

This polynomial is nonzero and vanishes for both possible values of $y - 2x$. □

Problem 35. (1996 Putnam). Let $[n] = \{1, 2, \dots, n\}$. A subset $X \subseteq [n]$ is called *selfish* if

$$|X| \in X.$$

A selfish set is minimal if none of its proper subsets is selfish. Determine the number of minimal selfish subsets of $[n]$.

Solution. Let X be a minimal selfish set and put $|X| = k$. Then $k \in X$. We claim that X contains no integer $j < k$. If it did, one could choose a j -element subset $Y \subsetneq X$ containing j , and then

$$|Y| = j \in Y,$$

so Y would be selfish, contradicting minimality.

Conversely, any k -element set that contains k and whose other elements are all greater than k is minimal selfish. Thus, for a fixed k , there are

$$\binom{n-k}{k-1}$$

choices. Summing over all possible k gives

$$\sum_{k \geq 1} \binom{n-k}{k-1} = \sum_{j \geq 0} \binom{n-j-1}{j} = F_n,$$

where $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$. Therefore the number of minimal selfish sets is

$$\boxed{F_n}.$$

□

Problem 36. (2002 AIME I). Let S be a set of distinct positive integers s.t.

$$1 \in S, \quad \max S = 2002.$$

Suppose that, for every $x \in S$, the arithmetic mean of the elements of $S \setminus \{x\}$ is an integer. Determine the maximum possible value of $|S|$.

Solution. Let

$$|S| = k, \quad T = \sum_{x \in S} x.$$

The hypothesis says that

$$T - x \equiv 0 \pmod{k-1}$$

for every $x \in S$. Hence all elements of S are congruent modulo $k-1$. Since $1, 2002 \in S$,

$$k-1 \mid 2001.$$

Put $d = k-1$. There must be at least $d+1$ distinct positive integers not exceeding 2002 that are congruent to 1 modulo d . The number of such integers is $2001/d + 1$, so

$$d+1 \leq \frac{2001}{d} + 1, \quad \text{hence} \quad d^2 \leq 2001.$$

Since

$$2001 = 3 \cdot 23 \cdot 29,$$

the largest divisor of 2001 not exceeding $\sqrt{2001}$ is 29. Therefore $k \leq 30$.

Equality is attainable. Take

$$S = \{1 + 29j \mid 0 \leq j \leq 28\} \cup \{2002\}.$$

This set has 30 elements, all congruent to 1 modulo 29. Its total is also congruent to 1 modulo 29, so deleting any one element leaves a sum divisible by 29. Thus the maximum is

$$\boxed{30}.$$

□

Problem 37. (1961 UTokyo Entrance Examination). In a triangle ABC , points $L \in \overline{BC}$, $M \in \overline{CA}$, and $N \in \overline{AB}$ satisfy

$$\frac{LC}{BL} = \frac{MA}{CM} = \frac{NB}{AN} = 2.$$

Let

$$P = AL \cap CN, \quad Q = AL \cap BM, \quad R = BM \cap CN.$$

Determine

$$\frac{[PQR]}{[ABC]}.$$

Solution. Area ratios are preserved by affine transformations, so take

$$A = (0, 0), \quad B = (1, 0), \quad C = (0, 1).$$

The division conditions give

$$L = \left(\frac{2}{3}, \frac{1}{3}\right), \quad M = \left(0, \frac{2}{3}\right), \quad N = \left(\frac{1}{3}, 0\right).$$

Solving the three pairs of line equations yields

$$P = \left(\frac{2}{7}, \frac{1}{7}\right), \quad Q = \left(\frac{4}{7}, \frac{2}{7}\right), \quad R = \left(\frac{1}{7}, \frac{4}{7}\right).$$

Therefore

$$[PQR] = \frac{1}{2} \left| \det \begin{pmatrix} 2/7 & 1/7 \\ -1/7 & 3/7 \end{pmatrix} \right| = \frac{1}{14}.$$

Since $[ABC] = 1/2$,

$$\boxed{\frac{[PQR]}{[ABC]} = \frac{1}{7}}.$$

□

Problem 38. (1995 KyotoU Entrance Examination). For a positive integer n , let $f(n)$ be the remainder when n is divided by 7, and define

$$g(n) = 3f\left(\sum_{k=1}^7 k^n\right).$$

Prove that

$$f(n^7) = f(n)$$

for every positive integer n . Then determine the maximum possible value of $g(n)$.

Solution. Fermat's little theorem gives

$$n^7 \equiv n \pmod{7}$$

for every integer n , including multiples of 7. Hence

$$f(n^7) = f(n).$$

Since f takes values in $\{0, 1, \dots, 6\}$, we have

$$g(n) \leq 18$$

for every positive integer n . This upper bound is attained at $n = 6$. Indeed, for $1 \leq k \leq 6$,

$$k^6 \equiv 1 \pmod{7},$$

while $7^6 \equiv 0 \pmod{7}$. Therefore

$$f\left(\sum_{k=1}^7 k^6\right) = 6,$$

and the maximum possible value is

$$\boxed{18}.$$

□

Problem 39. (1993 Tokyo Tech Entrance Examination). Let $P(x)$ be a polynomial of degree n . Suppose that

$$P(0), P(1), \dots, P(n)$$

are all integers. Prove that $P(m)$ is an integer for every $m \in \mathbb{Z}$.

Solution. We use induction on the degree. The assertion is immediate for constant polynomials. Let

$$Q(x) = P(x+1) - P(x).$$

Then Q has degree at most $n-1$, and

$$Q(0), Q(1), \dots, Q(n-1)$$

are integers. By the induction hypothesis,

$$Q(m) \in \mathbb{Z}$$

for every $m \in \mathbb{Z}$.

Since $P(0) \in \mathbb{Z}$, for $m > 0$ we have

$$P(m) = P(0) + \sum_{j=0}^{m-1} Q(j) \in \mathbb{Z}.$$

For $m < 0$,

$$P(m) = P(0) - \sum_{j=m}^{-1} Q(j) \in \mathbb{Z}.$$

Thus $P(m)$ is an integer for every integer m . □

Problem 40. (2009 Hitotsubashi Entrance Examination). Let $m, n \geq 2$ be integers satisfying

$$m^3 + 1 = n^3 + 10^3.$$

Determine m and n .

Solution. Rearranging and factoring gives

$$(m - n)(m^2 + mn + n^2) = 999.$$

Put $d = m - n > 0$. Then $d \mid 999$ and

$$3n^2 + 3dn + d^2 = \frac{999}{d}.$$

Since $n \geq 2$, the left-hand side exceeds d^2 , so $d^3 < 999$. Hence the only possible positive divisors are

$$d = 1, 3, 9.$$

Substitution gives no integer $n \geq 2$ for $d = 1$, gives $n = 9$ for $d = 3$, and gives only $n = 1$ or a negative value for $d = 9$. Therefore

$$\boxed{(m, n) = (12, 9)}.$$

The common value is the taxicab number

$$12^3 + 1^3 = 9^3 + 10^3 = 1729.$$

□

Problem 41. (2025 UTokyo Entrance Examination). For a positive integer a , define

$$f_a(x) = x^2 + x - a.$$

A square number means the square of a nonnegative integer. First prove that if n is a positive integer and $f_a(n)$ is a square number, then $n \leq a$.

Let N_a be the number of positive integers n for which $f_a(n)$ is a square number. Prove that

$$N_a = 1 \iff 4a + 1 \text{ is prime.}$$

Solution. Suppose that

$$n^2 + n - a = m^2$$

for some $m \in \mathbb{N}_0$. If $n \geq a + 1$, then

$$n^2 < n^2 + n - a < (n + 1)^2,$$

which is impossible for a square. Hence $n \leq a$.

Multiplying the equation by 4 and completing the square gives

$$(2n + 1 - 2m)(2n + 1 + 2m) = 4a + 1.$$

Both factors are positive odd integers. Conversely, let

$$uv = 4a + 1, \quad 1 \leq u \leq v,$$

with u and v odd. Since $uv \equiv 1 \pmod{4}$, the two factors are congruent modulo 4, and therefore

$$n = \frac{u + v - 2}{4}, \quad m = \frac{v - u}{4}$$

are nonnegative integers satisfying $f_a(n) = m^2$. This gives a bijection between positive factor pairs of $4a + 1$ and admissible values of n .

The factorization

$$4a + 1 = 1 \cdot (4a + 1)$$

always gives the solution $n = a$. There is exactly one factor pair if and only if $4a + 1$ is prime. Hence

$$\boxed{N_a = 1 \iff 4a + 1 \text{ is prime}}.$$

□

Problem 42. (1977 IMO). Let

$$f: \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$$

satisfy

$$f(n + 1) > f(f(n))$$

for every positive integer n . Prove that

$$f(n) = n$$

for every positive integer n .

Solution. We first prove, by induction on m , that

$$k \geq m \implies f(k) \geq m.$$

The assertion is immediate for $m = 1$. Suppose it holds for some m , and let $k \geq m + 1$. Write $k = n + 1$, so $n \geq m$. By the induction hypothesis,

$$f(n) \geq m,$$

and applying the same hypothesis once more gives

$$f(f(n)) \geq m.$$

Hence

$$f(k) = f(n + 1) > f(f(n)) \geq m,$$

so $f(k) \geq m + 1$. The induction is complete. Taking $k = m$ shows that

$$f(n) \geq n$$

for every n .

Applying this inequality with the input $f(n)$ gives

$$f(f(n)) \geq f(n).$$

Therefore

$$f(n+1) > f(f(n)) \geq f(n),$$

so f is strictly increasing.

If $f(n) > n$ for some n , then $f(n) \geq n+1$. Since f is increasing,

$$f(f(n)) \geq f(n+1),$$

contradicting the given inequality. Thus $f(n) \leq n$ for every n . Combined with $f(n) \geq n$, this yields

$$\boxed{f(n) = n}$$

for every positive integer n . □

Problem 43. (1985 IMO). Let n and k be relatively prime positive integers with $k < n$. Each element of

$$M = \{1, 2, \dots, n-1\}$$

is colored either blue or white. Suppose that

- (1) i and $n-i$ have the same color for every $i \in M$;
- (2) i and $|i-k|$ have the same color whenever $i \in M$ and $i \neq k$.

Prove that all elements of M have the same color.

Solution. For $1 \leq j \leq n-1$, let r_j be the least positive residue of jk modulo n . Since

$$\gcd(n, k) = 1,$$

the numbers

$$r_1, r_2, \dots, r_{n-1}$$

are precisely the elements of M in some order.

We show that consecutive terms in this list have the same color. Let $1 \leq j \leq n-2$. The equality $r_j + k = n$ cannot occur, because it would imply

$$(j+1)k \equiv 0 \pmod{n}$$

with $1 \leq j+1 < n$.

If $r_j + k < n$, then

$$r_{j+1} = r_j + k.$$

Since $r_{j+1} \neq k$, the second coloring condition gives

$$r_{j+1} \sim |r_{j+1} - k| = r_j,$$

where $x \sim y$ means that x and y have the same color.

If $r_j + k > n$, then

$$r_{j+1} = r_j + k - n.$$

The first condition gives $r_j \sim n - r_j$, and the second gives

$$n - r_j \sim |n - r_j - k| = r_j + k - n = r_{j+1}.$$

Thus $r_j \sim r_{j+1}$ in either case. Consequently all the residues $r_1, \dots, r_{n-1}$, and hence all elements of M , have the same color. □

Problem 44. (2001 IMO). Twenty-one girls and twenty-one boys took part in a mathematical competition. Each contestant solved at most six problems, and every girl–boy pair solved at least one problem in common. Prove that some problem was solved by at least three girls and at least three boys.

Solution. For each girl–boy pair, choose one problem that both contestants solved. Suppose, for contradiction, that no problem was solved by at least three girls and at least three boys.

Call a problem *girl-sparse* if it was solved by at most two girls. Fix a boy. The problems solved by this boy must collectively cover all 21 girls. Since he solved at most six problems, at least one of those problems was solved by at least three girls. Hence he solved at most five girl-sparse problems. Each such problem accounts for at most two of his 21 chosen girl–boy pairs. Therefore, for this boy, at most

$$5 \cdot 2 = 10$$

chosen pairs use a girl-sparse problem. Over all boys, there are at most

$$21 \cdot 10 = 210$$

such pairs.

By symmetry, at most 210 chosen pairs use a problem solved by at most two boys. Under the contrary assumption, every problem is girl-sparse or is solved by at most two boys. Thus all

$$21^2 = 441$$

chosen pairs would belong to the union of two sets containing at most 210 pairs each. This is impossible because

$$441 > 210 + 210.$$

Therefore some problem was solved by at least three girls and at least three boys. □

Problem 45. (2012 Putnam). Let

$$d_1, d_2, \dots, d_{12}$$

be real numbers in the open interval $(1, 12)$. Prove that there exist distinct indices i, j, k s.t. d_i, d_j, d_k are the side lengths of an acute triangle.

Solution. Reorder the numbers so that

$$d_1 \leq d_2 \leq \dots \leq d_{12}.$$

Suppose that no three of them are the side lengths of an acute triangle. For every $r \geq 3$, the three consecutive numbers d_{r-2}, d_{r-1}, d_r then satisfy

$$d_r^2 \geq d_{r-1}^2 + d_{r-2}^2.$$

Indeed, the reverse strict inequality would imply both the triangle inequality and, by the converse of the Pythagorean theorem, that the triangle is acute.

Let $F_1 = F_2 = 1$ and $F_r = F_{r-1} + F_{r-2}$. Since

$$d_1^2 > 1 = F_1, \quad d_2^2 > 1 = F_2,$$

induction gives

$$d_r^2 > F_r$$

for every r . In particular,

$$d_{12}^2 > F_{12} = 144,$$

so $d_{12} > 12$, contradicting $d_{12} \in (1, 12)$. Therefore the required three indices exist. □

Problem 46. (2017 Putnam). Let S be the smallest set of positive integers satisfying the following conditions:

- (1) $2 \in S$;
- (2) if $n^2 \in S$, then $n \in S$;
- (3) if $n \in S$, then $(n+5)^2 \in S$.

Determine all positive integers that do not belong to S .

Solution. Let

$$U = \{n \in \mathbb{Z}_{>0} \mid (n \neq 1) \wedge (5 \nmid n)\}.$$

The set U contains 2 and satisfies both closure conditions. Hence, by the minimality of S ,

$$S \subseteq U.$$

We prove the reverse inclusion by showing that every set T satisfying the three conditions contains U . Combining the second and third conditions gives

$$n \in T \implies n+5 \in T.$$

Thus T is closed under adding any nonnegative multiple of 5.

Starting from $2 \in T$, the following chain is valid:

$$2 \longrightarrow 49 \longrightarrow 54^2 \longrightarrow 56^2 \longrightarrow 56 \longrightarrow 121 \longrightarrow 11.$$

Here the first two arrows use the third condition, the arrows from 54^2 to 56^2 and from 56 to 121 use closure under adding multiples of 5, and the square-root arrows use the second condition. From $11 \in T$, closure under adding multiples of 5 gives

$$16 \in T, \quad 36 \in T.$$

Taking square roots gives $4, 6 \in T$. Since $4 \in T$, adding 5 gives $9 \in T$, and taking a square root gives $3 \in T$. Hence

$$2, 3, 4, 6 \in T.$$

Repeatedly adding 5 now shows that T contains every positive integer other than 1 and the positive multiples of 5. Thus $U \subseteq T$ for every admissible T , and in particular $U \subseteq S$.

Therefore the positive integers not in S are exactly

$$\boxed{1 \text{ and the positive multiples of } 5}.$$

□

Problem 47. (Fagnano's Problem). Let ABC be an acute triangle. Choose points

$$P \in \overline{BC}, \quad Q \in \overline{CA}, \quad R \in \overline{AB}.$$

Determine the triangle PQR whose perimeter

$$PQ + QR + RP$$

is as small as possible.

Solution. Reflect P across the lines CA and AB to points P_1 and P_2 , respectively. Since $Q \in CA$ and $R \in AB$,

$$PQ = P_1Q, \quad PR = P_2R.$$

Therefore

$$PQ + QR + RP = P_1Q + QR + RP_2 \geq P_1P_2.$$

Moreover,

$$AP_1 = AP_2 = AP, \quad \angle P_1AP_2 = 2\angle A,$$

so

$$P_1P_2 = 2AP \sin A.$$

This quantity is minimized when AP is the altitude from A to BC .

Let D, E, F be the feet of the altitudes from A, B, C , respectively. When $P = D$, the right-angle cyclic quadrilaterals determined by the altitude feet show that the reflections D_1, D_2 of D across CA and AB satisfy

$$D_1, E, F, D_2$$

collinear. Hence equality holds in the preceding triangle inequality for

$$(P, Q, R) = (D, E, F).$$

Thus the unique minimizing inscribed triangle is the orthic triangle of ABC .

□

Problem 48. (Fermat–Torricelli Problem). Given a triangle ABC , determine the point P in the plane for which

$$PA + PB + PC$$

is as small as possible.

Solution. First suppose that every angle of ABC is less than 120° . The standard construction with outward equilateral triangles produces a unique interior point P satisfying

$$\angle APB = \angle BPC = \angle CPA = 120^\circ.$$

Let $\mathbf{u}_A, \mathbf{u}_B, \mathbf{u}_C$ be the unit vectors from P toward A, B, C , respectively. Since these vectors are separated by angles of 120° ,

$$\mathbf{u}_A + \mathbf{u}_B + \mathbf{u}_C = \mathbf{0}.$$

For any point Q , let $\mathbf{q}$ be the vector from P to Q . The Cauchy–Schwarz inequality gives

$$QA \geq PA - \mathbf{u}_A \cdot \mathbf{q},$$

and similarly for B and C . Adding the three inequalities yields

$$QA + QB + QC \geq PA + PB + PC.$$

Thus this interior point is the minimizer.

Now suppose, without loss of generality, that

$$\angle BAC \geq 120^\circ.$$

Let $\mathbf{u}$ and $\mathbf{v}$ be the unit vectors from A toward B and C , and let $\mathbf{q}$ be the vector from A to an arbitrary point Q . Then

$$QB \geq AB - \mathbf{u} \cdot \mathbf{q}, \quad QC \geq AC - \mathbf{v} \cdot \mathbf{q}.$$

Since

$$\|\mathbf{u} + \mathbf{v}\| = 2 \cos \left(\frac{\angle BAC}{2} \right) \leq 1,$$

we also have

$$QA = \|\mathbf{q}\| \geq (\mathbf{u} + \mathbf{v}) \cdot \mathbf{q}.$$

Adding gives

$$QA + QB + QC \geq AB + AC,$$

with equality at $Q = A$. Therefore the minimizing point is

$\begin{cases} \text{the interior point making three } 120^\circ \text{ angles,} & \text{if all triangle angles are } < 120^\circ, \\ \text{the vertex whose angle is } \geq 120^\circ, & \text{otherwise.} \end{cases}$
--

□

Problem 49. (Napoleon's Theorem). On the three sides BC, CA, AB of an arbitrary triangle ABC , construct equilateral triangles externally. Let X, Y, Z be the centers of the equilateral triangles constructed on BC, CA, AB , respectively. Prove that XYZ is equilateral.

Solution. Identify the plane with $\mathbb{C}$, and let a, b, c be the complex coordinates of A, B, C , listed counterclockwise. Put

$$\rho = e^{-i\pi/3}.$$

The third vertices of the outward equilateral triangles on BC, CA, AB are

$$p = b + \rho(c - b), \quad q = c + \rho(a - c), \quad r = a + \rho(b - a),$$

respectively. Since the center of an equilateral triangle is its centroid, the complex coordinates x, y, z of X, Y, Z are

$$x = \frac{b + c + p}{3}, \quad y = \frac{c + a + q}{3}, \quad z = \frac{a + b + r}{3}.$$

Direct simplification gives

$$z - y = -\rho(y - x).$$

The complex number $-\rho$ has modulus 1 and argument $2\pi/3$. Hence the vector from Y to Z is obtained from the vector from X to Y by a rotation through 120° , without changing its length. Therefore

$$XY = YZ \quad \text{and} \quad \angle XYZ = 60^\circ,$$

so XYZ is equilateral.

□

Problem 50. (Butterfly Theorem). Let M be the midpoint of a chord PQ of a circle. Draw two other chords AB and CD through M , and define

$$X = AD \cap PQ, \quad Y = BC \cap PQ.$$

Prove that M is the midpoint of XY .

Solution. Choose Cartesian coordinates with

$$M = (0, 0)$$

and PQ as the x -axis. Since the perpendicular bisector of PQ passes through the center of the circle, its equation has the form

$$x^2 + (y - h)^2 = R^2.$$

Excluding only immediate degenerate cases, write the two chords through M as

$$AB : x = \alpha y, \quad CD : x = \beta y, \quad \alpha \neq \beta.$$

Let y_A, y_B, y_C, y_D be the y -coordinates of the corresponding points. Substitution into the circle equation and Viète's formulas give

$$\frac{y_A + y_B}{y_A y_B} = \frac{2h}{h^2 - R^2} = \frac{y_C + y_D}{y_C y_D}.$$

Write $X = (x_X, 0)$ and $Y = (x_Y, 0)$. Collinearity of A, X, D and of B, Y, C gives

$$x_X(y_A - y_D) = (\alpha - \beta)y_A y_D,$$

$$x_Y(y_B - y_C) = (\alpha - \beta)y_B y_C.$$

Hence

$$\frac{x_X + x_Y}{\alpha - \beta} = \frac{y_A y_B (y_C + y_D) - y_C y_D (y_A + y_B)}{(y_A - y_D)(y_B - y_C)} = 0.$$

Thus $x_X = -x_Y$, so X and Y are equidistant from the origin M . Therefore

$$\boxed{MX = MY}.$$

□

Problem 51. The number N of typhoons arriving in a given year is modeled by a Poisson distribution with parameter 1. Determine the probability that at least one typhoon arrives during the next year.

Solution. Since $N \sim \text{Pois}(1)$,

$$\mathbb{P}(N \geq 1) = 1 - \mathbb{P}(N = 0) = 1 - e^{-1}.$$

Therefore, the required probability is

$$\boxed{1 - \frac{1}{e}}.$$

□

Problem 52. Consider the functions

$$z^2, \quad \sin z, \quad \tan z, \quad \operatorname{Re}(z), \quad e^z.$$

Determine:

- (a) which functions are entire;
- (b) which function is meromorphic on $\mathbb{C}$ but not entire;
- (c) which function is not holomorphic on any nonempty open subset of $\mathbb{C}$.

Solution. The functions

$$z^2, \quad \sin z, \quad e^z$$

are entire. The function $\tan z$ is meromorphic but not entire because it has poles at

$$\frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

Finally, $\operatorname{Re}(z)$ is not complex differentiable at any point: its difference quotient equals 1 along real increments and 0 along purely imaginary increments. Thus the answers are

- (a) $z^2, \sin z, e^z$,
 (b) $\tan z$,
 (c) $\operatorname{Re}(z)$.

□

Problem 53. In the usual topology on $\mathbb{R}$, determine whether each statement is true or false.

- (1) There exists a perfect subset of $\mathbb{R}$ containing no nondegenerate interval.
- (2) The set $\mathbb{Z}$ is open.
- (3) The set $\mathbb{Z}$ is closed.
- (4) There exists a countable subset of $\mathbb{R}$ that is neither open nor closed.

Solution. The Cantor set is perfect and contains no nondegenerate interval, so (1) is true. The set $\mathbb{Z}$ is not open, because no open interval centered at an integer lies inside $\mathbb{Z}$, but it is closed because $\mathbb{R} \setminus \mathbb{Z}$ is open. Hence (2) is false and (3) is true. Finally, $\mathbb{Q}$ is countable and is neither open nor closed in $\mathbb{R}$, so (4) is true. Therefore,

$$(1) \text{ T}, \quad (2) \text{ F}, \quad (3) \text{ T}, \quad (4) \text{ T}.$$

□

Problem 54. Which of the following integers can be the order of a nonabelian simple group?

- (1) 60;
- (2) 48;
- (3) 30;
- (4) 77.

Solution. The alternating group A_5 is simple and has order 60, so 60 is possible.

A group of order $77 = 7 \cdot 11$ has a unique Sylow 11-subgroup, since

$$n_{11} \equiv 1 \pmod{11}, \quad n_{11} \mid 7,$$

so it is not simple. If a group of order 30 were simple, then it would have 6 Sylow 5-subgroups and 10 Sylow 3-subgroups. These would already contain

$$6(5 - 1) + 10(3 - 1) = 44$$

nonidentity elements, impossible in a group with only 29 nonidentity elements.

Finally, for a group of order 48, the number of Sylow 2-subgroups is 1 or 3. The first case gives a normal subgroup. In the second case, conjugation gives a homomorphism to S_3 ; it cannot be injective because $48 > 6$, so its kernel is a nontrivial proper normal subgroup. Hence the only possible order is

$$60.$$

□

Problem 55. Let

$$\mathbf{F}(x, y, z) = (z, x, y).$$

A plane through the origin intersects the sphere

$$x^2 + y^2 + z^2 = 4$$

in a great circle. Determine the plane for which the positively oriented circulation of $\mathbf{F}$ around this circle is as large as possible, and find the maximum circulation.

Solution. The curl is constant:

$$\nabla \times \mathbf{F} = (1, 1, 1).$$

Let $\mathbf{n}$ be the unit normal that determines the positive orientation of the circle. The circle bounds a disk of radius 2 and area 4π . By Stokes' Theorem,

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS = 4\pi(1, 1, 1) \cdot \mathbf{n}.$$

Cauchy–Schwarz gives

$$(1, 1, 1) \cdot \mathbf{n} \leq \sqrt{3},$$

with equality when

$$\mathbf{n} = \frac{1}{\sqrt{3}}(1, 1, 1).$$

Therefore the required plane and maximum circulation are

$$\boxed{x + y + z = 0}, \quad \boxed{4\pi\sqrt{3}}.$$

□

Problem 56. (Cantor's Diagonal Argument). Let $\{0, 1\}^{\mathbb{N}}$ be the set of all infinite binary sequences. Prove that no function

$$f: \mathbb{N} \rightarrow \{0, 1\}^{\mathbb{N}}$$

is surjective.

Solution. Write

$$f(n) = (a_{n1}, a_{n2}, a_{n3}, \dots), \quad a_{nk} \in \{0, 1\}.$$

Define a binary sequence $b = (b_1, b_2, \dots)$ by

$$b_n = 1 - a_{nn}.$$

For every $n \in \mathbb{N}$, the sequence b differs from $f(n)$ in its n th coordinate. Hence

$$b \neq f(n) \quad \text{for every } n \in \mathbb{N}.$$

Therefore b is not in the image of f , so f is not surjective.

□

Problem 57. On $\mathbb{R}^2$, determine which of the following functions is not a metric. For $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$, let

- (1) $d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$;
- (2) $d(\mathbf{x}, \mathbf{y}) = \min\{|x_1 - y_1|, |x_2 - y_2|\}$;
- (3) $d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2|$;
- (4) $d(\mathbf{x}, \mathbf{y}) = 0$ if $\mathbf{x} = \mathbf{y}$ and $d(\mathbf{x}, \mathbf{y}) = 1$ otherwise.

Solution. Choice (2) is not a metric. For example,

$$\mathbf{x} = (0, 0), \quad \mathbf{y} = (0, 1)$$

are distinct, but

$$d(\mathbf{x}, \mathbf{y}) = \min\{0, 1\} = 0.$$

Thus the identity-of-indiscernibles axiom fails. The other three functions are, respectively, the Euclidean, taxicab, and discrete metrics. Therefore, the answer is

$$\boxed{(2)}.$$

□

Problem 58. Which of the following sets has cardinality different from the other four?

- (1) the Cantor set;
- (2) the set of all functions $f: \mathbb{Q} \rightarrow \mathbb{R}$;
- (3) the set $C([0, 1])$ of all continuous functions $[0, 1] \rightarrow \mathbb{R}$;
- (4) $\mathbb{R}$;
- (5) the set of all algebraic numbers.

Solution. The Cantor set has cardinality $|\mathbb{R}| = \mathfrak{c}$. Also,

$$|\mathbb{R}^{\mathbb{Q}}| = \mathfrak{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = \mathfrak{c}.$$

A continuous function on $[0, 1]$ is determined by its values on the countable dense set $\mathbb{Q} \cap [0, 1]$, so

$$|C([0, 1])| \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c}.$$

Constant functions give the reverse inequality, hence $|C([0, 1])| = \mathfrak{c}$. By contrast, the algebraic numbers are countable because there are only countably many polynomials in $\mathbb{Z}[x]$, each with finitely many roots. Therefore the answer is

$$\boxed{\text{the set of all algebraic numbers}}.$$

□

Problem 59. Determine which of the following series converge:

- (1) $\sum_{n=1}^{\infty} \frac{1}{n}$;
- (2) $\sum_{\substack{p \text{ prime} \\ p+2 \text{ prime}}} \left(\frac{1}{p} + \frac{1}{p+2} \right)$;
- (3) $\sum_{n=1}^{\infty} \frac{1}{n^2}$;
- (4) $\sum_{n=1}^{\infty} \frac{1}{2^n}$.

Solution. The harmonic series in (1) diverges. Brun's Theorem from Mathematical Concepts states that the twin-prime reciprocal sum in (2) converges. The series in (3) is a p -series with $p = 2 > 1$, and the series in (4) is geometric with ratio $1/2$, so both converge. Hence

$$\boxed{(2), (3), \text{ and } (4)}.$$

□

Problem 60. A Menger sponge is obtained by repeatedly replacing each cube by 20 smaller cubes, each scaled by a factor of $1/3$. Determine its self-similar dimension.

Solution. If d is the self-similar dimension, Mathematical Concepts gives

$$20 \left(\frac{1}{3} \right)^d = 1.$$

Therefore

$$3^d = 20, \quad \boxed{d = \frac{\ln 20}{\ln 3}}.$$

□

Problem 61. Start with a tetrahedral polyhedron and cut seven pairwise disjoint tunnel-like through-holes through it. Let V , E , and F denote the numbers of vertices, edges, and faces in any polyhedral decomposition of the resulting boundary surface. Determine

$$V - E + F.$$

Solution. The original tetrahedral boundary is a sphere, which has genus 0 and Euler characteristic 2. Each disjoint through-hole adds one handle, so the resulting closed orientable surface has genus

$$g = 7.$$

By the Euler-characteristic formula from Mathematical Concepts,

$$V - E + F = 2 - 2g = 2 - 14.$$

Hence

$$\boxed{-12}.$$

□

Direct Calculation Problems

Problem 1. Finite Sums and Products.

Evaluate each of the following finite sums or products exactly. State any restrictions on the integer parameters that are needed for the displayed expression to be defined.

1.1. (Vandermonde's Identity). For $n \in \mathbb{N}_0$, determine

$$\sum_{k=0}^n \binom{n}{k}^2.$$

Solution. This is the $r = s = n$ specialization of Vandermonde's identity from Mathematical Concepts. Its coefficient interpretation gives the same result immediately: since

$$\binom{n}{k} = \binom{n}{n-k},$$

the required sum is the coefficient of x^n in

$$(1+x)^n(1+x)^n = (1+x)^{2n}.$$

Therefore,

$$\boxed{\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}}.$$

□

1.2. For $n \in \mathbb{N}_0$, determine

$$\sum_{k=0}^n (-1)^k \binom{n}{k} \frac{1}{k+1}.$$

Solution. Using

$$\frac{1}{k+1} = \int_0^1 x^k \, dx,$$

we obtain

$$\begin{aligned} \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{1}{k+1} &= \int_0^1 \sum_{k=0}^n \binom{n}{k} (-x)^k \, dx \\ &= \int_0^1 (1-x)^n \, dx \\ &= \boxed{\frac{1}{n+1}}. \end{aligned}$$

□

1.3. (Sine Product Formula). For an integer $n \geq 2$, determine

$$\prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right).$$

Solution. Let $\omega_n = e^{2\pi i/n}$. Factoring $z^n - 1$ and then dividing by $z - 1$ gives

$$\frac{z^n - 1}{z - 1} = \prod_{k=1}^{n-1} (z - \omega_n^k).$$

Letting $z \rightarrow 1$ and taking absolute values yields

$$n = \prod_{k=1}^{n-1} |1 - \omega_n^k| = \prod_{k=1}^{n-1} 2 \sin\left(\frac{k\pi}{n}\right).$$

Hence

$$\prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right) = \frac{n}{2^{n-1}}.$$

□

1.4. For an integer $n \geq 2$, determine

$$\prod_{k=2}^n \frac{k^3 - 1}{k^3 + 1}.$$

Solution. Factor each term as

$$\frac{k^3 - 1}{k^3 + 1} = \frac{k - 1}{k + 1} \frac{k^2 + k + 1}{k^2 - k + 1}.$$

The two products telescope separately:

$$\prod_{k=2}^n \frac{k - 1}{k + 1} = \frac{2}{n(n + 1)},$$

and, since $k^2 + k + 1 = (k + 1)^2 - (k + 1) + 1$,

$$\prod_{k=2}^n \frac{k^2 + k + 1}{k^2 - k + 1} = \frac{n^2 + n + 1}{3}.$$

Therefore,

$$\prod_{k=2}^n \frac{k^3 - 1}{k^3 + 1} = \frac{2(n^2 + n + 1)}{3n(n + 1)}.$$

□

1.5. For $m \in \mathbb{N}$ and $\sin x \neq 0$, determine

$$\prod_{k=0}^{m-1} \cos(2^k x).$$

Solution. Repeatedly apply

$$\sin(2t) = 2 \sin t \cos t.$$

This gives

$$\sin(2^m x) = 2^m \sin x \prod_{k=0}^{m-1} \cos(2^k x).$$

Since $\sin x \neq 0$,

$$\prod_{k=0}^{m-1} \cos(2^k x) = \frac{\sin(2^m x)}{2^m \sin x}.$$

□

1.6. For $n \in \mathbb{N}_0$, determine

$$\sum_{\substack{0 \leq k \leq n \\ 3|k}} \binom{n}{k}.$$

Solution. Let

$$\omega = e^{2\pi i/3}.$$

Since

$$1 + \omega^k + \omega^{2k} = \begin{cases} 3, & 3 \mid k, \\ 0, & 3 \nmid k, \end{cases}$$

the roots-of-unity filter gives

$$\begin{aligned} 3S_n &= \sum_{k=0}^n \binom{n}{k} (1 + \omega^k + \omega^{2k}) \\ &= 2^n + (1 + \omega)^n + (1 + \omega^2)^n. \end{aligned}$$

Now

$$1 + \omega = e^{i\pi/3}, \quad 1 + \omega^2 = e^{-i\pi/3}.$$

Therefore,

$$\boxed{S_n = \frac{2^n + 2 \cos(n\pi/3)}{3}}.$$

□

1.7. (1997 CSAT). For a permutation

$$(a_1, a_2, a_3, a_4)$$

of $\{1, 2, 3, 4\}$, let s_k be the number of entries to the right of a_k that are smaller than a_k , for $k = 1, 2, 3$. Define

$$|(a_1, a_2, a_3, a_4)| = s_1 + s_2 + s_3.$$

Determine the sum of this quantity over all 24 permutations.

Solution. For each unordered pair of distinct elements, exactly half of the 24 permutations place the larger element before the smaller one. Thus each of the

$$\binom{4}{2} = 6$$

pairs contributes one inversion in exactly 12 permutations. Hence the total inversion number is

$$\boxed{6 \cdot 12 = 72}.$$

□

Problem 2. Recurrence Relations.

Determine the requested term, closed form, or residue for each of the following recursively defined sequences.

2.1. (2006 VTRMC). Let $F_0 = 0$, $F_1 = 1$, and

$$F_{n+2} = F_{n+1} + F_n.$$

Determine the last decimal digit of F_{2006} .

Solution. Modulo 2, the Fibonacci sequence has period 3:

$$0, 1, 1, 0, 1, 1, \dots$$

Hence

$$F_{2006} \equiv F_2 \equiv 1 \pmod{2}.$$

Also

$$F_{n+5} = 5F_{n+1} + 3F_n,$$

so modulo 5,

$$F_{n+5} \equiv 3F_n.$$

Iterating four times gives $F_{n+20} \equiv F_n \pmod{5}$. Therefore

$$F_{2006} \equiv F_6 = 8 \equiv 3 \pmod{5}.$$

The unique decimal digit that is odd and congruent to 3 modulo 5 is

$$\boxed{3}.$$

□

2.2. Let $a_0 = 1$ and

$$a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right).$$

Determine a closed form for a_n .

Solution. Set

$$r_n = \frac{a_n - \sqrt{2}}{a_n + \sqrt{2}}.$$

Direct substitution into the recurrence gives

$$r_{n+1} = r_n^2.$$

Moreover,

$$r_0 = \frac{1 - \sqrt{2}}{1 + \sqrt{2}} = -(3 - 2\sqrt{2}).$$

Thus

$$r_n = (-(3 - 2\sqrt{2}))^{2^n}.$$

Solving the defining relation for a_n yields

$$\boxed{a_n = \sqrt{2} \frac{1 + r_n}{1 - r_n}, \quad r_n = (-(3 - 2\sqrt{2}))^{2^n}.$$

□

2.3. Let $a_0 = 0$ and

$$a_{n+1} = \frac{1 + a_n}{2 + a_n}.$$

Determine a closed form for a_n in terms of the Fibonacci sequence.

Solution. Let (F_n) be the Fibonacci sequence with $F_0 = 0$ and $F_1 = 1$. We claim that

$$a_n = \frac{F_{2n}}{F_{2n+1}}.$$

The formula holds for $n = 0$. If it holds for n , then

$$\begin{aligned} a_{n+1} &= \frac{F_{2n} + F_{2n+1}}{F_{2n} + 2F_{2n+1}} \\ &= \frac{F_{2n+2}}{F_{2n+3}}. \end{aligned}$$

Thus

$$a_n = \frac{F_{2n}}{F_{2n+1}}.$$

□

2.4. (2014 UTokyo Entrance Examination). A bag initially contains $a + 2$ white balls and one red ball, where a is a positive integer. After each draw, perform the following operation:

- (1) if a white ball is drawn, replace the contents by a white balls and one red ball;
- (2) if the red ball is drawn, remove it and leave the other balls unchanged.

Let p_n be the probability that the red ball is drawn on the n th draw. Determine p_n .

Solution. The first probability is

$$p_1 = \frac{1}{a + 3}.$$

If the red ball is drawn on the n th draw, it cannot be drawn next. If a white ball is drawn, the next state has $a + 1$ balls, exactly one of which is red. Therefore

$$p_{n+1} = \frac{1 - p_n}{a + 1}.$$

The fixed point of this recurrence is $1/(a + 2)$. Hence, with

$$q_n = p_n - \frac{1}{a + 2},$$

we have

$$q_{n+1} = -\frac{q_n}{a + 1}, \quad q_1 = -\frac{1}{(a + 3)(a + 2)}.$$

Consequently,

$$p_n = \frac{1}{a + 2} + \frac{(-1)^n}{(a + 3)(a + 2)(a + 1)^{n-1}}$$

for $n \geq 1$. In particular,

$$p_2 = \frac{a + 2}{(a + 3)(a + 1)}.$$

□

Problem 3. Limits.

Evaluate each of the following limits. In parameterized problems, determine the exact value for every parameter regime stated in the subproblem.

3.1. (2012 KAIST Problem of the Week). Let $a_0 = 3$ and

$$a_n = a_{n-1} + \sqrt{a_{n-1}^2 + 3}, \quad n \geq 1.$$

Determine

$$\lim_{n \rightarrow \infty} \frac{a_n}{2^n}.$$

Solution. Write

$$a_n = \sqrt{3} \cot \theta_n.$$

Since $a_0 = 3$, we may take $\theta_0 = \pi/6$. The half-angle identity

$$\cot \frac{\theta}{2} = \cot \theta + \csc \theta$$

transforms the recurrence into

$$a_n = \sqrt{3} (\cot \theta_{n-1} + \csc \theta_{n-1}) = \sqrt{3} \cot \frac{\theta_{n-1}}{2}.$$

Hence

$$\theta_n = \frac{\pi}{6 \cdot 2^n} \quad \text{and} \quad a_n = \sqrt{3} \cot \left(\frac{\pi}{6 \cdot 2^n} \right).$$

Using $t \cot t \rightarrow 1$ as $t \rightarrow 0$, we obtain

$$\boxed{\lim_{n \rightarrow \infty} \frac{a_n}{2^n} = \frac{6\sqrt{3}}{\pi}}.$$

□

3.2. (2013 KAIST Problem of the Week). For $a, b \in \mathbb{R}$, determine

$$\lim_{n \rightarrow \infty} n \left(1 - \frac{a}{n} - \frac{b \ln(n+1)}{n} \right)^n.$$

Solution. Put

$$c_n = a + b \ln(n+1).$$

Since $c_n/n \rightarrow 0$ and $c_n^2/n \rightarrow 0$, the logarithmic expansion gives

$$n \ln \left(1 - \frac{c_n}{n} \right) = -c_n + o(1).$$

Therefore,

$$n \left(1 - \frac{c_n}{n} \right)^n = e^{-a} n(n+1)^{-b} (1 + o(1)).$$

It follows that the limit is

$$\boxed{\begin{cases} \infty, & b < 1, \\ e^{-a}, & b = 1, \\ 0, & b > 1. \end{cases}}$$

□

3.3. (2014 KAIST Problem of the Week). For $x \geq 0$, let

$$f_n(x) = \frac{\prod_{k=1}^{n-1} (x+k)(x+k+1)}{(n!)^2}.$$

Determine

$$\lim_{n \rightarrow \infty} f_n(x).$$

Solution. The Gamma function gives

$$f_n(x) = \frac{\Gamma(n+x)\Gamma(n+x+1)}{\Gamma(x+1)\Gamma(x+2)\Gamma(n+1)^2}.$$

Using

$$\frac{\Gamma(n+\alpha)}{\Gamma(n+\beta)} \sim n^{\alpha-\beta},$$

we obtain

$$f_n(x) \sim \frac{n^{2x-1}}{\Gamma(x+1)\Gamma(x+2)}.$$

Consequently,

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0, & 0 \leq x < \frac{1}{2}, \\ \frac{8}{3\pi}, & x = \frac{1}{2}, \\ \infty, & x > \frac{1}{2}. \end{cases}$$

□

3.4. (2016 KAIST Problem of the Week). For $a \geq 0$, determine

$$\lim_{n \rightarrow \infty} n \int_{-1}^0 \left(x + \frac{x^2}{2} + e^{ax} \right)^n dx.$$

Solution. Let

$$g(x) = x + \frac{x^2}{2} + e^{ax}.$$

On $[-1, 0]$, we have $-1/2 \leq g(x) \leq 1$, with equality $g(x) = 1$ only at $x = 0$. Moreover,

$$g(0) = 1 \quad \text{and} \quad g'(0) = 1 + a.$$

The contribution from every interval $[-1, -\delta]$ is exponentially small. On the remaining interval, set $x = -t/n$. For each fixed $t \geq 0$,

$$g\left(-\frac{t}{n}\right)^n \longrightarrow e^{-(1+a)t}.$$

A uniform exponential bound near the endpoint permits dominated convergence, and therefore

$$\begin{aligned} \lim_{n \rightarrow \infty} n \int_{-1}^0 g(x)^n dx &= \int_0^\infty e^{-(1+a)t} dt \\ &= \boxed{\frac{1}{1+a}}. \end{aligned}$$

□

3.5. (2023 KAIST Problem of the Week). Determine

$$\lim_{n \rightarrow \infty} \left(\frac{\sum_{k=1}^{n+2} k^k}{\sum_{k=1}^{n+1} k^k} - \frac{\sum_{k=1}^{n+1} k^k}{\sum_{k=1}^n k^k} \right).$$

Solution. Let

$$S_n = \sum_{k=1}^n k^k.$$

The final two terms dominate the sum:

$$S_n = n^n \left(1 + \frac{1}{en} + O\left(\frac{1}{n^2}\right) \right).$$

Hence

$$\begin{aligned} \frac{S_{n+1}}{S_n} &= (n+1) \left(1 + \frac{1}{n} \right)^n \frac{1 + \frac{1}{e(n+1)} + O(n^{-2})}{1 + \frac{1}{en} + O(n^{-2})} \\ &= en + \frac{e}{2} + O\left(\frac{1}{n}\right). \end{aligned}$$

Replacing n by $n+1$ and subtracting gives

$$\boxed{\lim_{n \rightarrow \infty} \left(\frac{S_{n+2}}{S_{n+1}} - \frac{S_{n+1}}{S_n} \right) = e}.$$

□

Problem 4. Derivatives.

Compute each requested derivative or derivative-dependent quantity exactly. For high-order derivatives, a closed form involving standard sequences or harmonic numbers is acceptable.

4.1. (2006 HMMT). A nonzero polynomial $f \in \mathbb{R}[x]$ satisfies

$$f(x) = f'(x)f''(x).$$

Determine the leading coefficient of f .

Solution. Suppose that the leading term of f is cx^n , where $c \neq 0$. The leading term of $f'f''$ is

$$c^2 n^2 (n-1) x^{2n-3}.$$

Comparing degrees gives

$$n = 2n - 3,$$

so $n = 3$. Comparing leading coefficients then gives

$$c = 18c^2.$$

Since $c \neq 0$,

$$\boxed{c = \frac{1}{18}}.$$

□

4.2. Let

$$H_n = \sum_{k=1}^n \frac{1}{k}.$$

Determine

$$\left. \frac{d^{100}}{dx^{100}} (x^{100} \ln x) \right|_{x=1}.$$

Solution. For a real parameter α ,

$$\frac{d^{100}}{dx^{100}} x^\alpha = \alpha(\alpha-1) \cdots (\alpha-99) x^{\alpha-100}.$$

Differentiate both sides with respect to α and then set $\alpha = 100$ and $x = 1$. Since

$$\left. \frac{d}{d\alpha} \prod_{j=0}^{99} (\alpha - j) \right|_{\alpha=100} = 100! \sum_{k=1}^{100} \frac{1}{k},$$

we obtain

$$\boxed{100! H_{100}}.$$

□

Problem 5. Series.

Determine whether each of the following series converges. For every convergent series, determine its exact sum. In parameterized problems, state all parameter values for which convergence occurs.

5.1. (2016 KAIST Problem of the Week). Let k be a positive integer, and define

$$a_n = \begin{cases} 1, & (k+1) \nmid n, \\ -k, & (k+1) \mid n. \end{cases}$$

Determine

$$\sum_{n=1}^{\infty} \frac{a_n}{n}.$$

Solution. At the end of the N th complete block of length $k+1$, the partial sum is

$$\begin{aligned} \sum_{n=1}^{(k+1)N} \frac{a_n}{n} &= \sum_{n=1}^{(k+1)N} \frac{1}{n} - (k+1) \sum_{m=1}^N \frac{1}{(k+1)m} \\ &= H_{(k+1)N} - H_N. \end{aligned}$$

Since $H_N = \ln N + \gamma + o(1)$, these partial sums tend to $\ln(k+1)$. The terms inside the final incomplete block tend uniformly to zero, so the full series has the same limit. Therefore,

$$\boxed{\sum_{n=1}^{\infty} \frac{a_n}{n} = \ln(k+1)}.$$

□

5.2. (2017 KAIST Problem of the Week). Determine whether

$$\sum_{n=0}^{\infty} \frac{1}{4^n} \binom{2n}{n}$$

converges.

Solution. By the central binomial coefficient asymptotic,

$$\frac{1}{4^n} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}}.$$

Since the p -series $\sum n^{-1/2}$ diverges, the limit comparison test gives

$$\boxed{\sum_{n=0}^{\infty} \frac{1}{4^n} \binom{2n}{n} \text{ diverges.}}$$

□

5.3. (2014 KAIST Problem of the Week). Determine all positive integers ℓ for which

$$\sum_{n=1}^{\infty} \frac{n^3}{(n+1)(n+2) \cdots (n+\ell)}$$

converges, and compute its value whenever it converges.

Solution. The n th term is asymptotic to $n^{3-\ell}$, so the series converges exactly when

$$\ell \geq 5.$$

For such ℓ , the beta-integral identity gives

$$\frac{1}{(n+1)(n+2) \cdots (n+\ell)} = \frac{1}{(\ell-1)!} \int_0^1 x^n (1-x)^{\ell-1} dx.$$

Since

$$\sum_{n=1}^{\infty} n^3 x^n = \frac{x(1+4x+x^2)}{(1-x)^4},$$

term-by-term integration yields

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{n^3}{(n+1) \cdots (n+\ell)} &= \frac{1}{(\ell-1)!} \int_0^1 x(1+4x+x^2)(1-x)^{\ell-5} dx \\ &= \boxed{\frac{\ell(\ell+5)}{(\ell-4)(\ell-3)(\ell-2)(\ell-1)(\ell-1)!}}. \end{aligned}$$

□

5.4. Determine

$$\sum_{n=1}^{\infty} \frac{1}{n^2 \binom{2n}{n}}.$$

Solution. The Taylor expansion

$$2 \arcsin^2\left(\frac{x}{2}\right) = \sum_{n=1}^{\infty} \frac{x^{2n}}{n^2 \binom{2n}{n}}$$

holds for $|x| \leq 2$. It may be verified by differentiating both sides and using the power series for $1/\sqrt{1-x^2/4}$.

Setting $x = 1$ gives

$$\sum_{n=1}^{\infty} \frac{1}{n^2 \binom{2n}{n}} = 2 \arcsin^2\left(\frac{1}{2}\right) = 2 \left(\frac{\pi}{6}\right)^2 = \boxed{\frac{\pi^2}{18}}.$$

□

5.5. Evaluate

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}$$

in two ways.

Solution. First,

$$\frac{n}{(n+1)!} = \frac{1}{n!} - \frac{1}{(n+1)!},$$

so the series telescopes:

$$\sum_{n=1}^N \frac{n}{(n+1)!} = 1 - \frac{1}{(N+1)!} \rightarrow 1.$$

Alternatively, put $m = n + 1$ and use the exponential series:

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{n}{(n+1)!} &= \sum_{m=2}^{\infty} \frac{m-1}{m!} \\ &= \sum_{m=2}^{\infty} \frac{1}{(m-1)!} - \sum_{m=2}^{\infty} \frac{1}{m!} \\ &= (e-1) - (e-2) = 1. \end{aligned}$$

Thus both methods give

$$\boxed{1}.$$

□

5.6. (1998 CSAT). Let

$$a_1 = 1, \quad a_2 = 2, \quad a_{n+2} = a_{n+1} + a_n.$$

Determine

$$\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}a_{n+2}}.$$

Solution. The recurrence gives

$$\frac{a_n}{a_{n+1}a_{n+2}} = \frac{a_{n+2} - a_{n+1}}{a_{n+1}a_{n+2}} = \frac{1}{a_{n+1}} - \frac{1}{a_{n+2}}.$$

Thus the series telescopes:

$$\sum_{n=1}^N \frac{a_n}{a_{n+1}a_{n+2}} = \frac{1}{a_2} - \frac{1}{a_{N+2}}.$$

Since $a_n \rightarrow \infty$,

$$\boxed{\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}a_{n+2}} = \frac{1}{2}}.$$

□

Problem 6. Integrals.

Evaluate each of the following integrals. For indefinite integrals, the constant of integration may be omitted; for definite integrals, determine the exact value.

6.1. (2010 MIT Integration Bee). $\int_0^{\pi/2} \sin x \sin(2x) \sin(3x) \, dx$

Solution. By the product-to-sum identities,

$$\sin x \sin(3x) = \frac{1}{2}(\cos(2x) - \cos(4x)),$$

and hence

$$\sin x \sin(2x) \sin(3x) = \frac{1}{4}(\sin(2x) + \sin(4x) - \sin(6x)).$$

Therefore,

$$\int_0^{\pi/2} \sin x \sin(2x) \sin(3x) \, dx = \frac{1}{4} \left(1 - \frac{1}{3}\right) = \boxed{\frac{1}{6}}.$$

□

6.2. (2010 MIT Integration Bee). $\int_0^1 \sin^2(\ln x) \, dx$

Solution. Let $t = -\ln x$. Then $x = e^{-t}$ and $dx = -e^{-t} dt$, so

$$\int_0^1 \sin^2(\ln x) \, dx = \int_0^{\infty} e^{-t} \sin^2 t \, dt.$$

Since $\sin^2 t = (1 - \cos(2t))/2$,

$$\int_0^{\infty} e^{-t} \sin^2 t \, dt = \frac{1}{2} \left(1 - \frac{1}{5}\right) = \boxed{\frac{2}{5}}.$$

□

6.3. (2010 MIT Integration Bee). $\int_1^{\infty} \frac{dx}{x\sqrt{x^4-1}}$

Solution. Let $u = x^2$. Then

$$\frac{dx}{x} = \frac{du}{2u},$$

and therefore

$$\int_1^{\infty} \frac{dx}{x\sqrt{x^4-1}} = \frac{1}{2} \int_1^{\infty} \frac{du}{u\sqrt{u^2-1}}.$$

With $u = \sec \theta$, this becomes

$$\frac{1}{2} \int_0^{\pi/2} d\theta = \boxed{\frac{\pi}{4}}.$$

□

6.4. (2010 MIT Integration Bee). $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$

Solution. Let

$$I = \int_0^1 \frac{\ln(1+x)}{1+x^2} dx.$$

Substituting $x = \tan t$ gives

$$I = \int_0^{\pi/4} \ln(1 + \tan t) dt.$$

Under $t \mapsto \pi/4 - t$,

$$1 + \tan\left(\frac{\pi}{4} - t\right) = \frac{2}{1 + \tan t}.$$

Thus

$$I = \frac{\pi}{4} \ln 2 - I,$$

and consequently

$$\boxed{I = \frac{\pi}{8} \ln 2}.$$

□

6.5. (2010 MIT Integration Bee). $\int \frac{\cos x + x \sin x}{x(x + \cos x)} dx$

Solution. We recognize the logarithmic derivative

$$\frac{d}{dx}(\ln|x| - \ln|x + \cos x|) = \frac{\cos x + x \sin x}{x(x + \cos x)}.$$

Hence

$$\boxed{\int \frac{\cos x + x \sin x}{x(x + \cos x)} dx = \ln|x| - \ln|x + \cos x|}.$$

□

6.6. (2010 MIT Integration Bee). $\int_0^{\pi/2} \frac{dx}{\sin x + \sec x}$

Solution. Write

$$I = \int_0^{\pi/2} \frac{\cos x}{1 + \sin x \cos x} dx.$$

Replacing x by $\pi/2 - x$ also gives

$$I = \int_0^{\pi/2} \frac{\sin x}{1 + \sin x \cos x} dx.$$

Adding the two expressions and setting $u = \sin x - \cos x$, we obtain

$$2I = \int_{-1}^1 \frac{2}{3 - u^2} du.$$

Therefore,

$$\boxed{I = \frac{1}{\sqrt{3}} \ln(2 + \sqrt{3})}.$$

□

6.7. (2010 MIT Integration Bee). $\int_0^1 \sqrt{1+x} \sqrt{1+x} \sqrt{1+\cdots} dx$

Solution. For $x \in [0, 1]$, define the finite truncations by

$$R_1(x) = 1, \quad R_{n+1}(x) = \sqrt{1+xR_n(x)}.$$

Since $y \mapsto \sqrt{1+xy}$ is increasing for $x \in [0, 1]$, induction gives

$$1 \leq R_n(x) \leq \varphi \quad \text{and} \quad R_n(x) \leq R_{n+1}(x),$$

where $\varphi = (1+\sqrt{5})/2$ and $\varphi^2 = 1+\varphi$. Hence $R_n(x)$ converges pointwise to a limit $R(x)$. Passing to the limit in the recurrence gives

$$R(x)^2 = 1 + xR(x),$$

so positivity yields

$$R(x) = \frac{x + \sqrt{x^2 + 4}}{2}.$$

The uniform bound $0 \leq R_n(x) \leq \varphi$ permits integration by the dominated convergence theorem. Therefore,

$$\begin{aligned} \int_0^1 R(x) dx &= \frac{1}{2} \int_0^1 x dx + \frac{1}{2} \int_0^1 \sqrt{x^2 + 4} dx \\ &= \left[\frac{1 + \sqrt{5}}{4} + \ln \left(\frac{1 + \sqrt{5}}{2} \right) \right]. \end{aligned}$$

□

6.8. (2010 MIT Integration Bee). $\int \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2} \right) dx, \quad x > 0, \quad x \neq 1$

Solution. On any interval contained in $(0, \infty) \setminus \{1\}$,

$$\frac{d}{dx} \left(\frac{x}{\ln x} \right) = \frac{1}{\ln x} - \frac{1}{(\ln x)^2}.$$

Therefore,

$$\int \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2} \right) dx = \frac{x}{\ln x}.$$

□

6.9. (2010 MIT Integration Bee). $\int_1^2 \sqrt{x-1} \sqrt{2-x} dx$

Solution. Let $u = x - 1$. Then

$$\int_1^2 \sqrt{x-1} \sqrt{2-x} dx = \int_0^1 \sqrt{u(1-u)} du.$$

Setting $v = u - 1/2$ gives

$$\sqrt{u(1-u)} = \sqrt{\frac{1}{4} - v^2}.$$

Thus the integral is the area of a semicircle of radius $1/2$:

$$\left[\frac{\pi}{8} \right].$$

□

6.10. (2011 MIT Integration Bee). $\int \cos(\ln x) \, dx, \quad x > 0$

Solution. Let $t = \ln x$. Then $x = e^t$ and $dx = e^t dt$, so

$$\int \cos(\ln x) \, dx = \int e^t \cos t \, dt.$$

Therefore,

$$\boxed{\int \cos(\ln x) \, dx = \frac{x}{2} (\cos(\ln x) + \sin(\ln x))}.$$

□

6.11. (2011 MIT Integration Bee). $\int_0^\pi \cos x \cos(3x) \cos(5x) \, dx$

Solution. By repeated use of the product-to-sum identity,

$$\cos x \cos(3x) \cos(5x) = \frac{1}{4} (\cos x + \cos(3x) + \cos(7x) + \cos(9x)).$$

Each term has integral 0 on $[0, \pi]$. Hence

$$\boxed{\int_0^\pi \cos x \cos(3x) \cos(5x) \, dx = 0}.$$

□

6.12. (2011 MIT Integration Bee). $\int \sin(101x) \sin^{99} x \, dx$

Solution. Differentiating a suitable product gives

$$\begin{aligned} \frac{d}{dx} \left(\frac{1}{100} \sin(100x) \sin^{100} x \right) &= \sin^{99} x (\sin x \cos(100x) + \cos x \sin(100x)) \\ &= \sin(101x) \sin^{99} x. \end{aligned}$$

Therefore,

$$\boxed{\int \sin(101x) \sin^{99} x \, dx = \frac{1}{100} \sin(100x) \sin^{100} x}.$$

□

6.13. (2011 MIT Integration Bee). $\int x e^{e^{x^2} + x^2} \, dx$

Solution. Since

$$e^{e^{x^2} + x^2} = e^{e^{x^2}} e^{x^2},$$

let $u = e^{x^2}$. Then $du = 2xe^{x^2} dx$, and

$$\int x e^{e^{x^2} + x^2} \, dx = \frac{1}{2} \int e^u \, du.$$

Hence

$$\boxed{\int x e^{e^{x^2} + x^2} \, dx = \frac{1}{2} e^{e^{x^2}}}.$$

□

6.14. (2011 MIT Integration Bee). $\int \sqrt{\frac{1-x}{1+x}} dx, \quad -1 < x < 1$

Solution. For $-1 < x < 1$,

$$\sqrt{\frac{1-x}{1+x}} = \frac{1-x}{\sqrt{1-x^2}}.$$

Therefore,

$$\begin{aligned} \int \sqrt{\frac{1-x}{1+x}} dx &= \int \frac{dx}{\sqrt{1-x^2}} - \int \frac{x}{\sqrt{1-x^2}} dx \\ &= \boxed{\arcsin x + \sqrt{1-x^2}}. \end{aligned}$$

□

6.15. (Gabriel's Horn). Determine the volume and the area of the curved surface obtained by rotating

$$y = \frac{1}{x}, \quad x \geq 1,$$

about the x -axis.

Solution. The volume is

$$V = \pi \int_1^\infty \frac{1}{x^2} dx = \boxed{\pi}.$$

The curved surface area is

$$S = 2\pi \int_1^\infty \frac{1}{x} \sqrt{1 + \frac{1}{x^4}} dx.$$

Since the integrand is at least $1/x$, this integral dominates

$$2\pi \int_1^\infty \frac{1}{x} dx,$$

which diverges. Hence

$$\boxed{S = \infty}.$$

□

6.16. (Napkin Ring Problem). A cylindrical hole is drilled through the center of a sphere. The remaining solid has height 6. Determine its volume without being given either the sphere radius or the hole radius.

Solution. Let the original sphere radius be R and the cylindrical hole radius be r . If the remaining solid has height 6, then its half-height is 3, so

$$R^2 - r^2 = 9.$$

At height $z \in [-3, 3]$, the cross-section of the remaining solid is an annulus of area

$$\pi((R^2 - z^2) - r^2) = \pi(9 - z^2).$$

Therefore,

$$V = \pi \int_{-3}^3 (9 - z^2) dz = \boxed{36\pi}.$$

□

6.17. (Steinmetz Solid). Two infinite circular cylinders of radius R have perpendicular axes that intersect. Determine the volume of their intersection.

Solution. Choose coordinates so that the cylinders are

$$x^2 + y^2 \leq R^2, \quad x^2 + z^2 \leq R^2.$$

For fixed $x \in [-R, R]$, both y and z range independently over

$$\left[-\sqrt{R^2 - x^2}, \sqrt{R^2 - x^2}\right].$$

The cross-section is therefore a square of area

$$4(R^2 - x^2).$$

Hence

$$V = \int_{-R}^R 4(R^2 - x^2) \, dx = \boxed{\frac{16}{3}R^3}.$$

□

6.18. (Dirichlet Integral). Determine the improper integral

$$\int_0^\infty \frac{\sin x}{x} \, dx.$$

Solution. For $a > 0$, define

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} \, dx.$$

Differentiation under the integral sign gives

$$F'(a) = - \int_0^\infty e^{-ax} \sin x \, dx = -\frac{1}{1 + a^2}.$$

Also $F(a) \rightarrow 0$ as $a \rightarrow \infty$. Therefore,

$$F(a) = \int_a^\infty \frac{dt}{1 + t^2} = \frac{\pi}{2} - \arctan a.$$

Letting $a \rightarrow 0^+$ and applying Abel's limiting argument for the Dirichlet integral yields

$$\boxed{\int_0^\infty \frac{\sin x}{x} \, dx = \frac{\pi}{2}}.$$

□

6.19. (1944 Dalzell, On 22/7). Evaluate

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2} \, dx,$$

and use the result to prove that

$$\pi < \frac{22}{7}.$$

Solution. Polynomial division gives

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

Therefore,

$$\begin{aligned} I &= \int_0^1 (x^6 - 4x^5 + 5x^4 - 4x^2 + 4) dx - 4 \int_0^1 \frac{dx}{1+x^2} \\ &= \frac{22}{7} - \pi. \end{aligned}$$

Since the original integrand is positive on $(0, 1)$,

$$0 < I = \frac{22}{7} - \pi,$$

and hence

$$\boxed{\pi < \frac{22}{7}}.$$

□

6.20. (Borwein Integrals). For $n \in \mathbb{N}_0$, define

$$I_n = \int_0^\infty \prod_{k=0}^n \frac{\sin(x/(2k+1))}{x/(2k+1)} dx.$$

Determine how long the pattern $I_n = \pi/2$ persists, and show that the next case is smaller.

Solution. Put

$$a_k = \frac{1}{2k+1},$$

and let X_k be independent random variables uniformly distributed on $[-a_k, a_k]$. The characteristic function of X_k is

$$\frac{\sin(a_k t)}{a_k t}.$$

If $S_n = X_0 + \cdots + X_n$ has density f_n , Fourier inversion at the origin gives

$$I_n = \pi f_n(0).$$

Write $S_n = X_0 + Y_n$, where X_0 is uniform on $[-1, 1]$. If

$$\sum_{k=1}^n a_k \leq 1,$$

then $|Y_n| \leq 1$ always, and conditioning on Y_n gives

$$f_n(0) = \frac{1}{2}.$$

Thus $I_n = \pi/2$. Now

$$\sum_{k=1}^6 \frac{1}{2k+1} = \frac{43024}{45045} < 1,$$

whereas

$$\sum_{k=1}^7 \frac{1}{2k+1} = \frac{46027}{45045} > 1.$$

Hence

$$I_0 = I_1 = \cdots = I_6 = \frac{\pi}{2}.$$

For $n = 7$, the continuous random variable Y_7 has positive probability of satisfying $|Y_7| > 1$. Therefore

$$f_7(0) = \frac{1}{2}\mathbb{P}(|Y_7| \leq 1) < \frac{1}{2},$$

and consequently

$$I_7 < \frac{\pi}{2}.$$

□

6.21. (1987 Putnam). Determine

$$\int_2^4 \frac{\ln(9-x)}{\ln(9-x) + \ln(x+3)} dx.$$

Solution. Let

$$h(x) = \frac{\ln(9-x)}{\ln(9-x) + \ln(x+3)}.$$

The substitution $x \mapsto 6-x$ gives

$$h(6-x) = 1 - h(x).$$

Since the interval $[2, 4]$ is symmetric about 3,

$$2 \int_2^4 h(x) dx = \int_2^4 (h(x) + h(6-x)) dx = 2.$$

Therefore the integral is

$$\boxed{1}.$$

□

6.22. (1989 UTokyo Entrance Examination). Let

$$f(x) = \pi x^2 \sin(\pi x^2).$$

Determine the volume obtained by revolving the region bounded by $y = f(x)$, the x -axis, $x = 0$, and $x = 1$ about the y -axis.

Solution. Using cylindrical shells, the volume is

$$V = 2\pi \int_0^1 x f(x) dx = 2\pi^2 \int_0^1 x^3 \sin(\pi x^2) dx.$$

Let $u = x^2$. Then

$$V = \pi^2 \int_0^1 u \sin(\pi u) du.$$

Integration by parts gives

$$\int_0^1 u \sin(\pi u) du = \frac{1}{\pi}.$$

Hence

$$\boxed{V = \pi}.$$

□

6.23. (2014 CSAT). Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and odd. Suppose that

$$f(x) = \frac{\pi}{2} \int_1^{x+1} f(t) dt$$

for every real x , and that $f(1) = 1$. Determine

$$\pi^2 \int_0^1 x f(x+1) dx.$$

Solution. Differentiating the integral equation gives

$$f'(x) = \frac{\pi}{2} f(x+1),$$

so

$$f(x+1) = \frac{2}{\pi} f'(x).$$

Therefore the required expression is

$$2\pi \int_0^1 x f'(x) dx.$$

Integration by parts yields

$$2\pi \left([xf(x)]_0^1 - \int_0^1 f(x) dx \right) = 2\pi - 2\pi \int_0^1 f(x) dx.$$

Since f is odd and $f(1) = 1$, we have $f(-1) = -1$. Substituting $x = -1$ into the original equation gives

$$-1 = \frac{\pi}{2} \int_1^0 f(t) dt = -\frac{\pi}{2} \int_0^1 f(t) dt.$$

Thus

$$\int_0^1 f(t) dt = \frac{2}{\pi},$$

and the required value is

$$\boxed{2\pi - 4}.$$

□

6.24. For a real parameter $p > 1$, evaluate

$$\int_0^\infty \frac{dx}{1+x^p}.$$

Solution. Set

$$t = \frac{x^p}{1+x^p}.$$

Then

$$x^p = \frac{t}{1-t}$$

and the integral becomes

$$\frac{1}{p} \int_0^1 t^{1/p-1} (1-t)^{-1/p} dt = \frac{1}{p} B\left(\frac{1}{p}, 1 - \frac{1}{p}\right).$$

By the beta–gamma identity and Euler’s reflection formula,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)},$$

so

$$\boxed{\int_0^\infty \frac{dx}{1+x^p} = \frac{\pi}{p} \csc\left(\frac{\pi}{p}\right)}.$$

□

Problem 7. Complex Numbers.

Evaluate each of the following expressions exactly. When roots of unity occur, use $\omega_n = e^{2\pi i/n}$ unless another root is specified.

7.1. For an integer $n \geq 2$, determine

$$\sum_{k=0}^{n-1} (1 + \omega_n^k)^n.$$

Solution. Expand by the binomial theorem and interchange the finite sums:

$$\sum_{k=0}^{n-1} (1 + \omega_n^k)^n = \sum_{j=0}^n \binom{n}{j} \sum_{k=0}^{n-1} \omega_n^{kj}.$$

The inner sum is n when $n \mid j$ and 0 otherwise. For $0 \leq j \leq n$, only $j = 0$ and $j = n$ contribute. Hence

$$\boxed{\sum_{k=0}^{n-1} (1 + \omega_n^k)^n = 2n}.$$

□

7.2. (Quadratic Gauss Sum). Let $\omega = e^{2\pi i/5}$. Determine

$$\sum_{k=0}^4 \omega^{k^2}.$$

Solution. The quadratic residues modulo 5 are

$$0, 1, 4, 4, 1.$$

Therefore,

$$\sum_{k=0}^4 \omega^{k^2} = 1 + 2(\omega + \omega^{-1}) = 1 + 4 \cos\left(\frac{2\pi}{5}\right).$$

Since

$$2 \cos\left(\frac{2\pi}{5}\right) = \frac{\sqrt{5}-1}{2},$$

the sum is

$$\boxed{\sqrt{5}}.$$

□

7.3. For an integer $n \geq 2$, determine

$$\sum_{k=0}^{n-1} \frac{1}{2 - \omega_n^k}.$$

Solution. Let

$$p(z) = z^n - 1 = \prod_{k=0}^{n-1} (z - \omega_n^k).$$

Its logarithmic derivative is

$$\frac{p'(z)}{p(z)} = \sum_{k=0}^{n-1} \frac{1}{z - \omega_n^k}.$$

Setting $z = 2$ gives

$$\boxed{\sum_{k=0}^{n-1} \frac{1}{2 - \omega_n^k} = \frac{n2^{n-1}}{2^n - 1}}.$$

□

Problem 8. Polynomials and Roots.

Use the coefficients, factorization, or symmetric functions of the roots to compute each requested quantity exactly. Solving every polynomial explicitly is not required unless the problem asks for the roots.

8.1. (2006 HMMT). Determine the sum of the distinct real roots of

$$2x^6 - 3x^5 + 3x^4 + x^3 - 3x^2 + 3x - 1 = 0.$$

Solution. The polynomial factors as

$$2x^6 - 3x^5 + 3x^4 + x^3 - 3x^2 + 3x - 1 = (x^2 - x + 1)^2(2x^2 + x - 1).$$

The first factor has no real roots. The distinct real roots are therefore the roots of

$$2x^2 + x - 1 = (2x - 1)(x + 1),$$

namely $1/2$ and -1 . Their sum is

$$\boxed{-\frac{1}{2}}.$$

□

8.2. (1997 VTRMC). Two roots r_1, r_2 of

$$x^4 - x^3 + ax^2 - 8x - 8 = 0$$

satisfy $r_1 r_2 = 2$. Determine a , r_1 , and r_2 .

Solution. Since $r_1 r_2 = 2$, their quadratic factor has the form

$$x^2 + px + 2.$$

The remaining factor must have constant term -4 , so write

$$x^4 - x^3 + ax^2 - 8x - 8 = (x^2 + px + 2)(x^2 + qx - 4).$$

Comparing the coefficients of x^3 and x gives

$$p + q = -1, \quad 2q - 4p = -8.$$

Hence $p = 1$ and $q = -2$. Comparing the coefficient of x^2 gives $a = -4$. Thus

$$r_1, r_2 = \frac{-1 \pm i\sqrt{7}}{2}.$$

Therefore,

$$a = -4, \quad \{r_1, r_2\} = \left\{ \frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2} \right\}.$$

□

8.3. Let $\alpha_1, \dots, \alpha_5$ be the roots of

$$x^5 - 5x + 3 = 0.$$

Determine

$$\sum_{j=1}^5 (\alpha_j^4 + \alpha_j^5).$$

Solution. Let

$$s_k = \sum_{j=1}^5 \alpha_j^k.$$

Newton's identities for

$$x^5 + 0x^4 + 0x^3 + 0x^2 - 5x + 3$$

give

$$s_1 = s_2 = s_3 = 0,$$

$$s_4 + 4(-5) = 0, \quad s_5 + 5(3) = 0.$$

Hence $s_4 = 20$ and $s_5 = -15$, so

$$s_4 + s_5 = 5.$$

□

8.4. Let $\alpha_1, \dots, \alpha_4$ be the roots of

$$x^4 + x^3 + x^2 + x + 1 = 0.$$

Determine

$$\sum_{j=1}^4 \frac{1}{(1 - \alpha_j)^2}.$$

Solution. Let

$$p(x) = x^4 + x^3 + x^2 + x + 1 = \prod_{j=1}^4 (x - \alpha_j).$$

Then

$$\frac{p'(x)}{p(x)} = \sum_{j=1}^4 \frac{1}{x - \alpha_j}.$$

Differentiating gives

$$\sum_{j=1}^4 \frac{1}{(x - \alpha_j)^2} = \left(\frac{p'(x)}{p(x)} \right)^2 - \frac{p''(x)}{p(x)}.$$

Since

$$p(1) = 5, \quad p'(1) = 10, \quad p''(1) = 20,$$

the required sum is

$$\boxed{\left(\frac{10}{5} \right)^2 - \frac{20}{5} = 0}.$$

□

8.5. Let $\alpha_1, \alpha_2, \alpha_3$ be the roots of

$$x^3 - x - 1 = 0.$$

Determine

$$\prod_{j=1}^3 (1 + \alpha_j^2).$$

Solution. Let

$$p(x) = x^3 - x - 1 = \prod_{j=1}^3 (x - \alpha_j).$$

For each root α_j ,

$$(i - \alpha_j)(-i - \alpha_j) = 1 + \alpha_j^2.$$

Therefore,

$$\prod_{j=1}^3 (1 + \alpha_j^2) = p(i)p(-i).$$

Since

$$p(i) = -1 - 2i, \quad p(-i) = -1 + 2i,$$

we obtain

$$\boxed{p(i)p(-i) = 5}.$$

□

8.6. (1995 CSAT). Suppose that

$$\frac{1}{(x-1)(x-2)\cdots(x-10)} = \sum_{k=1}^{10} \frac{a_k}{x-k}$$

whenever the denominators are nonzero. Determine

$$a_1 + a_2 + \cdots + a_{10}.$$

Solution. Multiply the identity by x and let $x \rightarrow \infty$. The left-hand side tends to 0, while

$$\frac{x}{x-k} \rightarrow 1$$

for every k . Therefore

$$\boxed{a_1 + a_2 + \cdots + a_{10} = 0}.$$

□

8.7. Factor completely:

$$(a - b)^3 + (b - c)^3 + (c - a)^3.$$

Solution. Set

$$x = a - b, \quad y = b - c, \quad z = c - a.$$

Then $x + y + z = 0$. The identity

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$$

therefore gives

$$x^3 + y^3 + z^3 = 3xyz.$$

Hence

$$\boxed{3(a - b)(b - c)(c - a)}.$$

□

Problem 9. Determinants.

Compute each of the following determinants in exact form.

9.1. (2016 KAIST Problem of the Week). Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \begin{cases} 3, & i = j, \\ (-1)^{|i-j|}, & i \neq j. \end{cases}$$

Determine $\det(\mathbf{A})$.

Solution. Let

$$\mathbf{S} = \text{diag}((-1)^1, (-1)^2, \dots, (-1)^n).$$

Then $\det(\mathbf{S})^2 = 1$, and

$$\mathbf{SAS} = 2\mathbf{I}_n + \mathbf{J}_n,$$

where $\mathbf{J}_n$ is the all-ones matrix. The eigenvalues of $2\mathbf{I}_n + \mathbf{J}_n$ are $n + 2$ once and 2 with multiplicity $n - 1$. Therefore,

$$\boxed{\det(\mathbf{A}) = 2^{n-1}(n + 2)}.$$

□

9.2. (2016 KAIST Problem of the Week). Let T be a tree on the vertex set $\{1, 2, \dots, n\}$, and let d_{ij} be the distance between vertices i and j . Define

$$\mathbf{D}_T(x) = \left(x^{d_{ij}} \right)_{1 \leq i, j \leq n}.$$

Determine $\det(\mathbf{D}_T(x))$.

Solution. We use induction on n . The formula is immediate for $n = 1$. For $n \geq 2$, choose a leaf n whose unique neighbor is p . For every $j \neq n$,

$$d_{nj} = 1 + d_{pj}.$$

Apply the row operation

$$R_n \leftarrow R_n - xR_p.$$

The last row becomes zero except for its last entry $1 - x^2$. Next apply

$$C_n \leftarrow C_n - xC_p.$$

The last column also becomes zero except for the same last entry, while the remaining principal block is the matrix associated with the tree obtained by deleting the leaf. Thus

$$\det(\mathbf{D}_T(x)) = (1 - x^2) \det(\mathbf{D}_{T-n}(x)).$$

Induction gives

$$\boxed{\det(\mathbf{D}_T(x)) = (1 - x^2)^{n-1}}.$$

□

9.3. (2015 KAIST Problem of the Week). For $n \in \mathbb{N}$ and $x \in \mathbb{R}$, let $\mathbf{T}_n(x) = (t_{ij})$ be the $n \times n$ tridiagonal matrix defined by

$$t_{ij} = \begin{cases} 1, & i = j, \\ x, & |i - j| = 1, \\ 0, & |i - j| \geq 2. \end{cases}$$

Determine $\det(\mathbf{T}_n(x))$.

Solution. Put

$$D_n(x) = \det(\mathbf{T}_n(x)), \quad D_0(x) = 1.$$

Expansion along the final row gives

$$D_n(x) = D_{n-1}(x) - x^2 D_{n-2}(x), \quad D_0(x) = D_1(x) = 1.$$

The polynomial

$$\sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{n-k}{k} x^{2k}$$

has these initial values and satisfies the same recurrence by Pascal's identity. Therefore,

$$\boxed{\det(\mathbf{T}_n(x)) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{n-k}{k} x^{2k}}.$$

Equivalently, if $x \neq \pm 1/2$ and

$$r_{\pm} = \frac{1 \pm \sqrt{1 - 4x^2}}{2},$$

then

$$D_n(x) = \frac{r_+^{n+1} - r_-^{n+1}}{r_+ - r_-};$$

for $x = \pm 1/2$, this becomes $D_n(x) = (n+1)/2^n$.

□

9.4. (2011 KAIST Problem of the Week). Let $\mathbf{M} = (m_{ij})$ be the $n \times n$ matrix defined by

$$m_{ij} = i(i+1)(i+2) \cdots (i+j-2),$$

where the empty product m_{i1} is 1. Determine $\det(\mathbf{M})$.

Solution. For $j \geq 1$, define the monic polynomial

$$p_{j-1}(t) = t(t+1) \cdots (t+j-2),$$

with $p_0(t) = 1$. Then

$$m_{ij} = p_{j-1}(i).$$

Replacing the monomial basis $1, t, t^2, \dots, t^{n-1}$ by the monic basis $p_0, p_1, \dots, p_{n-1}$ uses a unit upper-triangular change-of-basis matrix. Hence $\det(\mathbf{M})$ equals the Vandermonde determinant at $1, 2, \dots, n$:

$$\begin{aligned} \det(\mathbf{M}) &= \prod_{1 \leq i < j \leq n} (j - i) \\ &= \prod_{k=1}^{n-1} k!. \end{aligned}$$

□

9.5. (2012 KAIST Problem of the Week). List all nonempty subsets of $\{1, 2, \dots, n\}$ as $S_1, S_2, \dots, S_{2^n-1}$, and define the $(2^n - 1) \times (2^n - 1)$ matrix $\mathbf{A} = (a_{ij})$ by

$$a_{ij} = \begin{cases} 1, & S_i \cap S_j \neq \emptyset, \\ 0, & S_i \cap S_j = \emptyset. \end{cases}$$

Determine $|\det(\mathbf{A})|$.

Solution. Index rows and columns by the nonempty subsets of $[n]$. Define

$$c_{S,U} = \begin{cases} 1, & U \subseteq S, \\ 0, & U \not\subseteq S, \end{cases}$$

and let $\mathbf{C} = (c_{S,U})$. Also let $\mathbf{E}$ be diagonal with

$$e_{U,U} = (-1)^{|U|+1}.$$

Inclusion-exclusion gives

$$(\mathbf{CEC}^\top)_{S,T} = \sum_{\emptyset \neq U \subseteq S \cap T} (-1)^{|U|+1} = \begin{cases} 1, & S \cap T \neq \emptyset, \\ 0, & S \cap T = \emptyset. \end{cases}$$

Thus

$$\mathbf{A} = \mathbf{CEC}^\top.$$

Order the subsets by increasing cardinality. Then $\mathbf{C}$ is triangular with every diagonal entry equal to 1, so $\det(\mathbf{C}) = 1$. Since every diagonal entry of $\mathbf{E}$ is ± 1 ,

$$|\det(\mathbf{A})| = 1.$$

□

Problem 10. Eigenvalues and Matrix Powers.

Determine the requested eigenvalues or matrix powers exactly. Use diagonalization, polynomial identities, or invariant subspaces whenever they make the computation shorter than direct multiplication.

10.1. (1997 VTRMC). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}.$$

Determine the $(1, 1)$ -entry of $\mathbf{A}^{100}$.

Solution. The eigenvalues of $\mathbf{A}$ are

$$\lambda_{\pm} = 1 \pm \sqrt{3}.$$

The matrix is diagonalizable, and the two spectral projections have equal $(1, 1)$ -entries $1/2$. Hence

$$(\mathbf{A}^{100})_{11} = \frac{\lambda_+^{100} + \lambda_-^{100}}{2}.$$

Therefore,

$$\boxed{(\mathbf{A}^{100})_{11} = \frac{(1 + \sqrt{3})^{100} + (1 - \sqrt{3})^{100}}{2}}.$$

□

10.2. (Fibonacci Q-Matrix). Let

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Determine $\mathbf{Q}^n$ for $n \in \mathbb{N}$ in terms of the Fibonacci sequence.

Solution. Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. We claim that

$$\mathbf{Q}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$$

for $n \geq 1$. The formula is true for $n = 1$, and multiplying the displayed matrix by $\mathbf{Q}$ advances every entry according to the Fibonacci recurrence. Thus

$$\boxed{\mathbf{Q}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}}.$$

□

10.3. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Determine $\mathbf{A}^n$ for $n \in \mathbb{N}_0$.

Solution. Write

$$\mathbf{A} = \mathbf{I}_3 + \mathbf{N}, \quad \mathbf{N} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since $\mathbf{N}^3 = \mathbf{0}$, the nilpotent-matrix recognition rule and the binomial theorem from Mathematical Concepts give

$$\mathbf{A}^n = \mathbf{I}_3 + n\mathbf{N} + \binom{n}{2}\mathbf{N}^2.$$

Therefore,

$$\mathbf{A}^n = \begin{pmatrix} 1 & n & \binom{n}{2} \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}.$$

□

10.4. Determine the eigenvalues of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Solution. Write

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}.$$

This is a rank-one outer product. Its only possible nonzero eigenvalue is

$$\begin{pmatrix} 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 6.$$

Since the matrix has rank 1, the remaining two eigenvalues are 0. Therefore,

$$\boxed{6, 0, 0}.$$

□

10.5. Determine the eigenvalues of

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 3 & 0 \\ -2 & 3 & 0 & 1 \\ 3 & 0 & 1 & -2 \\ 0 & 1 & -2 & 3 \end{pmatrix}.$$

Solution. Reorder the coordinates as $(1, 3, 2, 4)$. The resulting similar matrix has the block form

$$\begin{pmatrix} \mathbf{B} & -2\mathbf{I}_2 \\ -2\mathbf{I}_2 & \mathbf{C} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

The vectors $(1, 1)$ and $(1, -1)$ simultaneously diagonalize $\mathbf{B}$ and $\mathbf{C}$. On the corresponding two-dimensional invariant subspaces, the matrix reduces respectively to

$$\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -2 & -2 \\ -2 & 2 \end{pmatrix}.$$

Their eigenvalues are $6, 2$ and $\pm 2\sqrt{2}$. Hence

$$\boxed{6, 2, 2\sqrt{2}, -2\sqrt{2}}.$$

□

10.6. Let $\mathbf{A}$ be the 2022×2022 matrix whose diagonal entries are 0 and whose off-diagonal entries are 1. Determine all eigenvalues, including multiplicities.

Solution. We have

$$\mathbf{A} = \mathbf{J}_{2022} - \mathbf{I}_{2022}.$$

The all-ones vector is an eigenvector of $\mathbf{J}_{2022}$ with eigenvalue 2022, while every vector whose coordinates sum to zero has eigenvalue 0. Subtracting the identity shifts all eigenvalues by -1 . Therefore,

$$\boxed{2021 \text{ with multiplicity } 1, \quad -1 \text{ with multiplicity } 2021}.$$

□

10.7. Determine all eigenvalues of

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Solution. The matrix cyclically shifts the coordinates and satisfies

$$\mathbf{P}^4 = \mathbf{I}_4.$$

Hence every eigenvalue λ satisfies $\lambda^4 = 1$. The characteristic polynomial is

$$\lambda^4 - 1,$$

whose roots are

$$1, \quad -1, \quad i, \quad -i.$$

Therefore the eigenvalues are

$$\boxed{1, -1, i, -i}.$$

□

10.8. Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix

$$a_{ij} = ij, \quad 1 \leq i, j \leq n.$$

Determine the sum of all eigenvalues of $\mathbf{A}$, counted with algebraic multiplicity.

Solution. Mathematical Concepts gives that the sum of the eigenvalues, counted with algebraic multiplicity, equals the trace. Therefore

$$\sum_{k=1}^n \lambda_k = \text{tr}(\mathbf{A}) = \sum_{i=1}^n i^2 = \boxed{\frac{n(n+1)(2n+1)}{6}}.$$

Equivalently, $\mathbf{A} = \mathbf{u}\mathbf{u}^T$ for $\mathbf{u} = (1, 2, \dots, n)^T$, so $\mathbf{A}$ has rank 1 and its unique nonzero eigenvalue is $\|\mathbf{u}\|^2$.

□

10.9. Let

$$\mathbf{A} = \begin{pmatrix} -11 & 4 \\ -30 & 11 \end{pmatrix}.$$

Determine $\mathbf{A}^{2016}$.

Solution. A direct multiplication gives

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}_2.$$

Hence

$$\mathbf{A}^{2016} = (\mathbf{A}^2)^{1008} = \boxed{\mathbf{I}_2}.$$

□

Memorization Problems

Problem 1. The Clay Mathematics Institute's Yang–Mills Existence and Mass Gap problem asks for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^d$ with a positive mass gap. Determine d , and state whether this case has been solved.

Solution. As recorded in Facts to Memorize, the Millennium Prize Problem is posed on four-dimensional Euclidean space:

$$\boxed{d = 4}.$$

As of 2026, the required rigorous construction and proof of a positive mass gap in this dimension remain open. □

Problem 2. John Milnor first exhibited exotic spheres in dimension n : smooth manifolds homeomorphic, but not diffeomorphic, to the standard sphere S^n . Later work showed that the group Θ_n of oriented diffeomorphism classes of homotopy n -spheres has exactly k elements in this dimension. Determine

$$n + k.$$

Solution. Milnor's first exotic spheres were seven-dimensional, and Kervaire–Milnor theory gives

$$\Theta_7 \cong \mathbb{Z}/28\mathbb{Z}.$$

Thus, with orientation taken into account, there are 28 classes in total: one standard class and 27 exotic classes. Therefore $n = 7$ and $k = 28$, so

$$\boxed{n + k = 35}.$$

□

Problem 3. This public-key cryptosystem was developed in 1977 by Ronald Rivest, Adi Shamir, and Leonard Adleman. It supports public-key encryption and digital signatures, and its standard key construction uses a modulus formed from two large primes. The three researchers received the 2002 ACM A. M. Turing Award for their contribution to practical public-key cryptography. Identify the cryptosystem.

Solution. The cryptosystem is

$$\boxed{\text{RSA}}.$$

Its name consists of the initials of Rivest, Shamir, and Adleman. □

Problem 4. For a connected finite graph, Euler's criterion states that an Euler circuit exists exactly when every vertex has even degree, while an Euler trail with distinct endpoints exists exactly when precisely two vertices have a certain degree parity. What kind of vertices occur zero times in the first case and exactly twice in the second?

Solution. They are

$$\boxed{\text{vertices of odd degree}}.$$

□

Problem 5. This Norwegian mathematician lived from 1802 to 1829 and died at the age of 26. Despite poverty and illness, he made fundamental contributions to analysis and algebra. He is especially known for proving that the general quintic equation cannot be solved by radicals. Identify the mathematician.

Solution. The mathematician is

Niels Henrik Abel

.

□

Problem 6. This closed surface has no boundary and is nonorientable, so it has no globally consistent distinction between two sides. Unlike a Möbius strip, it is closed. It cannot be embedded in three-dimensional Euclidean space, although it can be represented there by a self-intersecting immersion. Identify the surface.

Solution. The surface is the

Klein bottle

.

□

Problem 7. This German mathematician was born near Königsberg in 1862, held a professorship at the University of Königsberg, and moved to the University of Göttingen in 1895. His work shaped algebra, number theory, analysis, and the axiomatic foundations of geometry, and his *Grundlagen der Geometrie* appeared in 1899. Identify the mathematician.

Solution. The mathematician is

David Hilbert

.

□

Problem 8. In game theory, this is a strategy profile in which no player can improve their payoff by changing only their own strategy while all other players keep theirs fixed. It is named after the mathematician portrayed in the film *A Beautiful Mind*. What is this concept?

Solution. The concept is a

Nash equilibrium

.

□

Problem 9. A complex number is called this if it is not a root of any nonzero polynomial with rational coefficients. The numbers π and e are standard examples. What is such a number called?

Solution. It is called a

transcendental number

.

□

Problem 10. A positive integer is called this if it equals the sum of its positive proper divisors. For example,

$$6 = 1 + 2 + 3, \quad 28 = 1 + 2 + 4 + 7 + 14.$$

The Euclid–Euler theorem states that every even integer of this type has the form

$$2^{p-1}(2^p - 1),$$

where $2^p - 1$ is prime. What is such an integer called?

Solution. It is called a

perfect number

.

□

Problem 11. An Euler path traverses every edge of a graph exactly once. What is the standard name for a path that visits every vertex of a graph exactly once?

Solution. It is called a

Hamiltonian path

.

□

Problem 12. The obverse of the Fields Medal depicts a mathematician from antiquity, while the reverse depicts a sphere inscribed in a cylinder. Identify the mathematician and state the volume ratio of the sphere to the circumscribed cylinder.

Solution. The mathematician is Archimedes. If the sphere has radius r , then the cylinder has radius r and height $2r$, so

$$\frac{V_{\text{sphere}}}{V_{\text{cylinder}}} = \frac{\frac{4}{3}\pi r^3}{2\pi r^3} = \frac{2}{3}.$$

Thus the answer is

Archimedes and $2 : 3$

.

□

Problem 13. Match the four 2018 Fields Medalists with the following principal mathematical anchors:

- (1) p -adic geometry;
- (2) boundedness of Fano varieties and the BAB conjecture;
- (3) optimal transport;
- (4) number theory and homogeneous dynamics.

Solution. The Fields Medal table and laureate anchors in Facts to Memorize give the matching

Peter Scholze $\longleftrightarrow$ p -adic geometry,
 Caucher Birkar $\longleftrightarrow$ Fano varieties and BAB,
 Alessio Figalli $\longleftrightarrow$ optimal transport,
 Akshay Venkatesh $\longleftrightarrow$ number theory and homogeneous dynamics.

□

Problem 14. Maryna Viazovska proved optimal sphere-packing results in dimensions 8 and 24. What is the classical three-dimensional sphere-packing conjecture called, and who proved it?

Solution. The three-dimensional statement is the

Kepler conjecture

,

proved by

Thomas Hales

.

The later Flyspeck project formally verified the proof.

□

Problem 15. Which of the following is not one of the combinatorial log-concavity conjectures associated with June Huh's work?

- (1) Rota's conjecture;
- (2) Mason's conjecture;
- (3) Dowling–Wilson conjecture;
- (4) Sullivan conjecture.

Solution. The first three belong to the circle of matroid and combinatorial log-concavity results associated with June Huh and his collaborators. The Sullivan conjecture belongs to algebraic topology. Therefore, the odd one out is

Sullivan conjecture.

□

Problem 16. Identify the Abel Prize laureate or laureates for each year:

2018, 2019, 2020, 2021, 2022.

Solution. The sequence is

2018 : Robert Langlands,
 2019 : Karen Uhlenbeck,
 2020 : Hillel Furstenberg and Gregory Margulis,
 2021 : László Lovász and Avi Wigderson,
 2022 : Dennis Sullivan.

□

Problem 17. Give the Fields Medalists conventionally associated with the following national firsts:

Brazil, Japan, Iran, Vietnam.

Solution. The standard associations are

Brazil $\longleftrightarrow$ Artur Avila,
 Japan $\longleftrightarrow$ Kunihiko Kodaira,
 Iran $\longleftrightarrow$ Maryam Mirzakhani,
 Vietnam $\longleftrightarrow$ Ngô Bảo Châu.

□

Problem 18. Which of the following mathematicians was born earliest?

- (1) Guillaume de l'Hôpital;
- (2) Isaac Newton;
- (3) Gottfried Wilhelm Leibniz;
- (4) Colin Maclaurin.

Solution. Newton was born in 1643 under the modern Gregorian calendar convention, earlier than Leibniz in 1646, l'Hôpital in 1661, and Maclaurin in 1698. Therefore the answer is

Isaac Newton

.

□

Problem 19. What is Hilbert's first problem in the list of twenty-three problems presented in 1900?

Solution. Hilbert's first problem asks for the resolution of the

continuum hypothesis

.

In modern set theory, Gödel and Cohen showed that the continuum hypothesis is independent of ZFC, assuming ZFC is consistent.

□

Problem 20. Match each number system with the mathematician most directly associated with its introduction or early systematic development. The mathematicians are Hamilton, Hensel, Hewitt, and Conway.

- (1) quaternions;
- (2) p -adic numbers;
- (3) hyperreal numbers;
- (4) surreal numbers.

Solution. The standard associations are

quaternions $\longleftrightarrow$ Hamilton, p -adic numbers $\longleftrightarrow$ Hensel, hyperreal numbers $\longleftrightarrow$ Hewitt, surreal numbers $\longleftrightarrow$ Conway.
--

Robinson later made hyperreal methods central to nonstandard analysis, and Knuth introduced the name *surreal numbers*.

□

Problem 21. Match each clue with the corresponding paradox.

- (a) Among 23 people, a shared birthday has probability greater than $1/2$.
- (b) Alternating two losing games can create a winning game.
- (c) A ball can be reassembled into two congruent copies using finitely many nonmeasurable pieces.
- (d) Dominance reasoning and expected utility reasoning recommend different choices when a highly accurate predictor has already filled two boxes.

Solution. The matching is

(a) birthday paradox, (b) Parrondo's paradox, (c) Banach–Tarski paradox, (d) Newcomb's paradox.
--

□

Problem 22. Euclid's fifth postulate is often replaced in modern elementary geometry by the statement that through a point outside a given line there is exactly one parallel line. What is this equivalent statement called?

Solution. It is

Playfair's axiom.

□

Problem 23. Match the three constant-curvature geometries with the number of parallels through a point outside a given line and with the angle sum of a triangle.

- (a) Euclidean geometry;
- (b) hyperbolic geometry;
- (c) elliptic or spherical geometry.

Solution. The matching is

(a) Euclidean: exactly one parallel, angle sum $= \pi$,
 (b) hyperbolic: infinitely many disjoint lines, angle sum $< \pi$,
 (c) elliptic / spherical: no disjoint geodesic line, angle sum $> \pi$.

□

Problem 24. Match each foundational clue with the corresponding mathematician or pair of mathematicians.

- (1) diagonal argument and the comparison of infinite cardinalities;
- (2) the paradox of the set of all sets that do not contain themselves;
- (3) the axiomatic system ZF, with ZFC obtained by adding the Axiom of Choice;
- (4) the independence of the Continuum Hypothesis from ZFC, assuming consistency.

Solution. The correct associations are

(1) Georg Cantor,
 (2) Bertrand Russell,
 (3) Ernst Zermelo and Abraham Fraenkel,
 (4) Kurt Gödel and Paul Cohen.

□

Problem 25. Arrange the following classes of commutative rings from strongest to weakest:

- (1) unique factorization domain;
- (2) field;
- (3) integral domain;
- (4) principal ideal domain;
- (5) Euclidean domain.

Solution. The standard implication chain is

field $\implies$ Euclidean domain $\implies$ PID $\implies$ UFD $\implies$ integral domain.

□

Problem 26. Identify the Joseon mathematician and chief state councillor who wrote *Gusuryak*, exhibited orthogonal Latin squares of order 9, and devised Jisuguimundo, also called the Hexagonal Tortoise Problem.

Solution. The mathematician is

Choi Seok-jeong.

□

Problem 27. Identify the Oxford mathematics lecturer Charles Lutwidge Dodgson by the pen name under which he wrote *Alice's Adventures in Wonderland*.

Solution. His pen name was

Lewis Carroll.

□

Problem 28. Who received the first Abel Prize, awarded in 2003?

Solution. The first Abel laureate was

Jean-Pierre Serre.

□

Problem 29. Identify the mathematician associated with the modularity conjecture formerly called the Taniyama–Shimura conjecture, a central ingredient in the proof of Fermat's Last Theorem.

Solution. The mathematician is

Goro Shimura.

□

Problem 30. What collective pseudonym was adopted by the group of mainly French mathematicians who sought a unified axiomatic presentation of modern mathematics?

Solution. The collective pseudonym is

Nicolas Bourbaki.

□

Problem 31. Brook Taylor published the general expansion now bearing his name in the early eighteenth century, but which Scottish mathematician had already found the underlying series method in 1671?

Solution. The mathematician was

James Gregory.

□

Problem 32. Which French mathematician is remembered for the theorem asserting an interior zero of the derivative when a differentiable function has equal endpoint values, and also adopted the now-standard indexed radical notation for n th roots?

Solution. The mathematician is

Michel Rolle

.

□

Problem 33. The rule commonly called l'Hôpital's rule appeared in l'Hôpital's calculus textbook, but the underlying method came from lessons by another mathematician. Identify that mathematician.

Solution. The mathematician was

Johann Bernoulli

.

□

Problem 34. In mathematics, the abbreviation FTA commonly refers to two different fundamental theorems. Name both.

Solution. They are

the Fundamental Theorem of Algebra and the Fundamental Theorem of Arithmetic

.

□

Problem 35. What is the standard name for the ring

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}?$$

Solution. Its elements are the

Gaussian integers

.

□

Problem 36. The standard mathematical model of single-fold origami is usually described by seven axioms named after two mathematicians. What are these axioms commonly called?

Solution. They are the

Huzita–Hatori axioms

.

The historical naming is not a strict priority claim: Jacques Justin had published all seven axioms earlier, while Humiaki Huzita and Koshiro Hatori are associated with the commonly used name.

□

Problem 37. What branch of mathematics studies objects together with morphisms between them, with associative composition and identity morphisms?

Solution. The branch is

category theory

.

□

Problem 38. In the title of al-Khwarizmi's foundational algebra treatise, two Arabic terms describe restoring a subtracted quantity and balancing or reducing corresponding quantities. Name the two terms.

Solution. The terms are

al-jabr and *al-muqabala*.

The word *algebra* derives from *al-jabr*. □

Problem 39. How many reachable configurations does the standard $3 \times 3 \times 3$ Rubik's Cube have?

Solution. The corner pieces can be permuted in $8!$ ways and oriented in 3^7 independent ways. The edges can be permuted in $12!$ ways and oriented in 2^{11} ways, but the corner and edge permutation parities must agree, giving a factor of $1/2$. Thus

$$8! 3^7 12! 2^{10} = \boxed{43\,252\,003\,274\,489\,856\,000}.$$

□

Problem 40. For a regular space curve, what is the standard collective name for the data

$$\{\mathbf{T}, \mathbf{N}, \mathbf{B}, \kappa, \tau\},$$

consisting of the unit tangent, principal normal, binormal, curvature, and torsion?

Solution. This collection is the

Frenet–Serret apparatus.

□

Problem 41. The median identity

$$AB^2 + AC^2 = 2(AM^2 + BM^2),$$

where M is the midpoint of BC , is sometimes called the Pappus median theorem in the Republic of Korea and Japan. What is its standard international name?

Solution. It is

Apollonius' theorem.

□

Problem 42. A fifteenth family of convex pentagons capable of tiling the plane was found in 2015. By 2017, an exhaustive computer-assisted search had settled the classification. As of 2026, how many families of convex pentagons that tile the plane are known to exist in this classification?

Solution. The exhaustive classification found no additional families beyond the previously known fifteen. Therefore the number is

15.

□

Reasoning and Puzzle Problems

Problem 1. (Sum-and-Product Puzzle). Two integers x and y satisfy

$$1 < x < y, \quad x + y \leq 100.$$

The sum $x + y$ is disclosed privately to S , and the product xy is disclosed privately to P . The preceding information is common knowledge. The following conversation then takes place:

- (1) P : “I do not know the pair.”
- (2) S : “I knew that you did not know the pair.”
- (3) P : “Now I know the pair.”
- (4) S : “Now I also know the pair.”

Determine (x, y) .

Solution. Let

$$\Omega_0 = \{(x, y) \in \mathbb{N}^2 \mid (1 < x < y) \wedge (x + y \leq 100)\}.$$

For $\Omega \subseteq \Omega_0$ and $\omega = (x, y) \in \Omega$, define the information sets

$$\mathcal{I}_S^\Omega(\omega) = \{(u, v) \in \Omega \mid u + v = x + y\},$$

$$\mathcal{I}_P^\Omega(\omega) = \{(u, v) \in \Omega \mid uv = xy\}.$$

The four announcements successively restrict the state space as follows:

$$\Omega_1 = \{\omega \in \Omega_0 \mid |\mathcal{I}_P^{\Omega_0}(\omega)| > 1\},$$

$$\Omega_2 = \{\omega \in \Omega_1 \mid \mathcal{I}_S^{\Omega_0}(\omega) \subseteq \Omega_1\},$$

$$\Omega_3 = \{\omega \in \Omega_2 \mid |\mathcal{I}_P^{\Omega_2}(\omega)| = 1\},$$

and

$$\Omega_4 = \{\omega \in \Omega_3 \mid |\mathcal{I}_S^{\Omega_3}(\omega)| = 1\}.$$

The finite elimination is summarized by

stage	public information retained	number of states
0	$(1 < x < y) \wedge (x + y \leq 100)$	2352
1	P does not know the pair	1747
2	S knew that P did not know	145
3	P now knows the pair	86
4	S now knows the pair	1

At the second stage, the possible sums are exactly

$$11, 17, 23, 27, 29, 35, 37, 41, 47, 53.$$

These values follow by checking, for each admissible sum, whether every compatible factorization has at least two representations in Ω_0 . After the third announcement, the only sum-information set of cardinality 1 is

$$\mathcal{I}_S^{\Omega_3}(4, 13) = \{(4, 13)\}.$$

Therefore,

$$\Omega_4 = \{(4, 13)\},$$

and the unique pair is

$$\boxed{(x, y) = (4, 13)}.$$

□

Problem 2. (Bertrand's Box Paradox). Three boxes contain coins as follows:

$$GG, \quad GS, \quad SS,$$

where G denotes a gold coin and S denotes a silver coin. A box is chosen uniformly at random, and then one of its two coins is chosen uniformly at random. The chosen coin is gold. Determine the conditional probability that the other coin in the same box is also gold.

Solution. Let B_{GG} be the event that the chosen box is GG , and let G be the event that the observed coin is gold. Then

$$\mathbb{P}(G \mid B_{GG}) = 1, \quad \mathbb{P}(G \mid B_{GS}) = \frac{1}{2}, \quad \mathbb{P}(G \mid B_{SS}) = 0.$$

Since the three boxes are initially equally likely, Bayes' theorem gives

$$\begin{aligned} \mathbb{P}(B_{GG} \mid G) &= \frac{\mathbb{P}(G \mid B_{GG})\mathbb{P}(B_{GG})}{\mathbb{P}(G \mid B_{GG})\mathbb{P}(B_{GG}) + \mathbb{P}(G \mid B_{GS})\mathbb{P}(B_{GS})} \\ &= \frac{1 \cdot (1/3)}{1 \cdot (1/3) + (1/2) \cdot (1/3)} = \boxed{\frac{2}{3}}. \end{aligned}$$

Equivalently, after observing gold, the selected coin is equally likely to be any of the three gold coins present among the boxes. Two of those three coins lie in the GG box. □

Problem 3. (Penney's Game). The first player chooses a pattern of three coin tosses, such as HHT. After seeing this choice, the second player chooses a different pattern of length 3. A fair coin is then tossed repeatedly until one of the two patterns first appears as three consecutive outcomes; the player who chose that pattern wins.

Prove that the second player can always choose a pattern whose winning probability is greater than $1/2$.

Solution. For two patterns X and Y of length 3, define the overlap score

$$C(X, Y) = \sum_{\substack{1 \leq k \leq 3 \\ \text{suffix}_k(X) = \text{prefix}_k(Y)}} 2^k.$$

A standard fair-betting argument gives the probability that Y occurs before X :

$$\mathbb{P}(Y \text{ before } X) = \frac{C(X, X) - C(X, Y)}{C(X, X) - C(X, Y) + C(Y, Y) - C(Y, X)}.$$

Indeed, start a new unit-stake gambler before each toss, with one team betting successively on X and the other on Y , doubling its capital after each correct fair bet. At the stopping time, the surviving capital of the team betting on Y is $C(X, Y)$ if X wins and $C(Y, Y)$ if Y wins; the analogous statement holds for the other team. Equality of the two expected net gains yields the displayed formula.

Write the first player's pattern as

$$X = x_1x_2x_3,$$

and let $\overline{x_2}$ denote the opposite face of x_2 . The second player chooses

$$Y = \overline{x_2}x_1x_2.$$

By interchanging heads and tails if necessary, it is enough to check the following four cases:

X	Y	$\mathbb{P}(Y \text{ before } X)$
HHH	THH	$7/8$
HHT	THH	$3/4$
HTH	HHT	$2/3$
HTT	HHT	$2/3$

These values follow immediately from the overlap formula. Each is strictly greater than $1/2$, so the second player always has the required choice. $\square$

Problem 4. (Josephus Problem). The people numbered

$$1, 2, \dots, n$$

stand in a circle. Starting with person 1, count repeatedly around the circle and eliminate every second person. After each elimination, resume counting with the next remaining person. Determine the number of the last survivor.

Solution. Let $J(n)$ denote the number of the survivor. If $n = 2m$, the first circuit eliminates all even-numbered people. Relabeling the survivors

$$1, 3, \dots, 2m - 1$$

as $1, 2, \dots, m$ gives

$$J(2m) = 2J(m) - 1.$$

A similar relabeling when $n = 2m + 1$ gives

$$J(2m + 1) = 2J(m) + 1.$$

Write

$$n = 2^r + \ell, \quad 0 \leq \ell < 2^r.$$

Starting from $J(2^r) = 1$, a strong induction using these recurrences yields

$$\boxed{J(n) = 2\ell + 1}.$$

In binary notation this has a particularly simple interpretation. If

$$n = (1b_{r-1}b_{r-2} \cdots b_0)_2,$$

then

$$J(n) = (b_{r-1}b_{r-2} \cdots b_01)_2;$$

the leading binary digit of n is moved to the end. $\square$

Problem 5. (Conway's Soldiers). On an infinite square grid, every square on or below a fixed horizontal line initially contains a soldier, and every square strictly above the line is empty. A legal move jumps one soldier horizontally or vertically over an adjacent soldier into the next square, which must be empty; the jumped soldier is removed. Prove that no soldier can reach the fifth row above the initial line.

Solution. Fix a target square T in the fifth row and put

$$q = \frac{\sqrt{5} - 1}{2}, \quad q + q^2 = 1.$$

Give each square Q the weight

$$w(Q) = q^{d(Q,T)},$$

where d is the Manhattan distance, and let W be the sum of the weights of all occupied squares.

Consider three consecutive squares in the direction of a legal jump. Their distances from T have one of the forms

$$(m+2, m+1, m), \quad (m, m+1, m+2), \quad (m+1, m, m+1).$$

In the first case, the jump moves toward T and

$$q^{m+2} + q^{m+1} = q^m$$

because $q+q^2 = 1$. In the second case, $q^m + q^{m+1} > q^{m+2}$, and in the third case, $q^{m+1} + q^m > q^{m+1}$. Thus no legal move increases W .

Take the initial boundary to be $y = 0$ and $T = (0, 5)$. The initial weight is

$$\begin{aligned} W_0 &= \sum_{m=0}^{\infty} \sum_{k \in \mathbb{Z}} q^{|k|+m+5} \\ &= q^5 \left(1 + 2 \sum_{k=1}^{\infty} q^k \right) \left(\sum_{m=0}^{\infty} q^m \right) = 1. \end{aligned}$$

Before a soldier could first reach T , only finitely many moves would have occurred. Hence only finitely many initially occupied squares could have been altered, so at least one untouched soldier of positive weight would remain. If T were occupied, it alone would contribute weight 1, and the untouched soldier would contribute an additional positive amount. This would give $W > 1$, contradicting $W \leq W_0 = 1$.

Therefore no soldier can reach the fifth row. □

Problem 6. (Three Utilities Problem). Three houses and three utility companies—water, gas, and electricity—are represented by six distinct points in the plane. Is it possible to join every house to every utility company by curves so that no two curves intersect and no curve passes through any of the six points other than its endpoints? Prove the answer.

Solution. Such a drawing would be a planar embedding of the complete bipartite graph $K_{3,3}$. This graph has

$$V = 6, \quad E = 9.$$

The graph $K_{3,3}$ is simple, bipartite, and has no bridges. Thus every face boundary in a planar embedding would be a closed walk of length at least 4. If F is the number of faces, double counting edge–face incidences gives

$$2E \geq 4F, \quad \text{so} \quad F \leq \frac{E}{2}.$$

Euler's formula would then give

$$2 = V - E + F \leq V - E + \frac{E}{2},$$

and hence

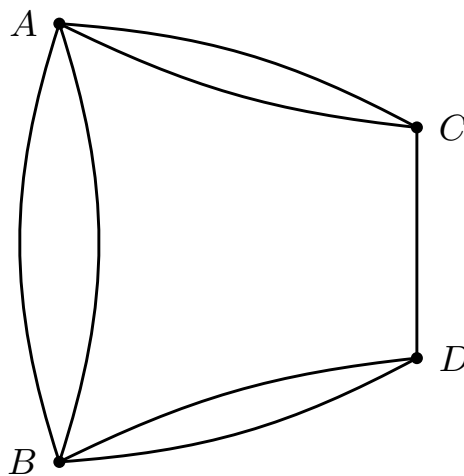
$$E \leq 2V - 4.$$

For $K_{3,3}$ this would require

$$9 \leq 2 \cdot 6 - 4 = 8,$$

which is impossible. Thus the requested system of nonintersecting connections cannot be drawn in the plane. □

Problem 7. (Seven Bridges of Königsberg). The city of Königsberg was divided by the Pregel River into four land masses, connected by seven bridges as shown below. Is it possible to start at one land mass, cross each bridge exactly once, and end at some land mass? Prove your answer.



Solution. Replace each land mass by a vertex and each bridge by an edge. Then the problem becomes the following graph-theoretic question: does the resulting connected multigraph have an Euler trail, that is, a trail using every edge exactly once?

In the graph above, the four vertex degrees are

$$3, \quad 3, \quad 3, \quad 5.$$

In particular, all four vertices have odd degree.

A connected graph has an Euler trail if and only if it has exactly zero or two vertices of odd degree. Since this graph has four odd vertices, an Euler trail does not exist.

Therefore it is impossible to cross each of the seven bridges exactly once. □

Problem 8. (15 Puzzle). In the standard fifteen puzzle, a legal move consists of sliding one tile adjacent to the blank square into the blank square. Starting from the solved position on the left, determine whether one can reach the position on the right, in which only the tiles 14 and 15 have been interchanged.

solved position

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

14–15 challenge

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

Solution. Read the tiles row by row from left to right and top to bottom, omitting the blank square. Let I be the number of inversions in this resulting list, and let r be the row number of the blank square counted from the top. We claim that the parity of

$$I + r$$

is invariant under every legal move.

If the blank square moves horizontally, then the order of the numbered tiles in the row-by-row list does not change, so I is unchanged, and r is also unchanged. Hence the parity of $I + r$ is unchanged.

If the blank square moves vertically, then one tile moves up or down by one row. In the row-by-row list, that tile passes across exactly three other tiles, so the inversion number changes by 3, and therefore its parity changes. At the same time, the blank square moves to an adjacent row, so the parity of r also changes. Hence the parity of $I + r$ remains unchanged in this case as well.

In the solved position, the inversion number is

$$I = 0,$$

and the blank square lies in row 4, so

$$I + r \equiv 0 + 4 \equiv 0 \pmod{2}.$$

In the target position, the list differs only by the transposition of 14 and 15, so

$$I = 1,$$

while the blank square is still in row 4. Thus

$$I + r \equiv 1 + 4 \equiv 1 \pmod{2}.$$

Since the parity of $I + r$ is invariant, the target position cannot be reached from the solved position.

Therefore the 14–15 challenge is impossible.

□

Problem 9. (Chomp). An $m \times n$ rectangular chocolate bar consists of unit-square pieces. The lower-left piece is poisoned.

On each turn, a player chooses one remaining piece and eats that piece together with every remaining piece weakly above it and weakly to its right. The player who eats the poisoned piece loses.

Prove that whenever the initial bar contains more than the poisoned piece alone, the first player has a winning strategy.

Solution. Since the game is finite and has no draws, one of the two players has a winning strategy. Suppose, for contradiction, that the second player has one.

Let the first player eat only the upper-right corner piece. According to the supposed winning strategy, the second player must have a non-poisoned response; otherwise the second player is forced to eat the poisoned piece and loses. Let that response be a remaining piece C . Eating C also removes the upper-right corner, because that corner lies above and to the right of C . Hence the position obtained after these two moves is exactly the position that would have resulted had the first player chosen C on the very first move.

After the original two-move sequence, this position is losing for the player whose turn it is, namely the first player. The first player could instead begin by choosing C and hand the identical losing position to the second player. This contradicts the assumption that the second player has a winning strategy.

Therefore the first player has a winning strategy. The argument proves its existence without identifying the winning first move in general. □

Problem 10. (Braess's Paradox). A total traffic flow of 4000 units travels from S to T in the directed network with routes

$$S \rightarrow A \rightarrow T \quad \text{and} \quad S \rightarrow B \rightarrow T.$$

If x denotes the flow on a road, its travel time is given by

Road	$S \rightarrow A$	$A \rightarrow T$	$S \rightarrow B$	$B \rightarrow T$
Travel time	$x/100$	45	45	$x/100$

Each driver chooses a route of minimum travel time, taking the total road flows as given.

Determine the equilibrium travel time in the original network. Then a new directed road $A \rightarrow B$ with travel time 0 is added. Determine the new equilibrium travel time.

Solution. Let y units use the upper route $S \rightarrow A \rightarrow T$. Then $4000 - y$ units use the lower route, and their travel times are

$$C_U(y) = \frac{y}{100} + 45, \quad C_L(y) = 45 + \frac{4000 - y}{100}.$$

At equilibrium both routes are used and have equal travel time. Thus

$$\frac{y}{100} + 45 = 45 + \frac{4000 - y}{100},$$

so $y = 2000$ and

$$C_U = C_L = 65.$$

After the road $A \rightarrow B$ is added, let f , g , and h be the flows on the routes

$$S \rightarrow A \rightarrow T, \quad S \rightarrow B \rightarrow T, \quad S \rightarrow A \rightarrow B \rightarrow T,$$

respectively. Put

$$x = f + h, \quad z = g + h,$$

for the flows on $S \rightarrow A$ and $B \rightarrow T$. The three route costs are

$$C_U = \frac{x}{100} + 45, \quad C_L = 45 + \frac{z}{100}, \quad C_M = \frac{x+z}{100}.$$

If $f > 0$, equilibrium would require $C_U \leq C_M$, which implies $z \geq 4500$, impossible because the total flow is 4000. Hence $f = 0$. Similarly, $g = 0$. Therefore

$$h = 4000, \quad x = z = 4000,$$

and every driver uses the new middle route with travel time

$$C_M = 40 + 40 = 80.$$

Thus adding the zero-time road increases the equilibrium travel time from

$$\boxed{65} \quad \text{to} \quad \boxed{80}.$$

□

Problem 11. (100 Prisoners and 100 Boxes). One hundred prisoners and one hundred boxes are labeled $1, 2, \dots, 100$. The prisoners' labels are placed in the boxes according to a uniformly random permutation.

The prisoners enter the room one at a time. Each prisoner may open at most 50 boxes and must restore them before leaving. They may agree on a strategy beforehand but may not communicate after the process begins. They all survive if and only if every prisoner finds their own label.

Find a strategy whose probability of saving all the prisoners is greater than 30%.

Solution. Regard the contents of the boxes as a permutation π of $\{1, 2, \dots, 100\}$, where box i contains the label $\pi(i)$. Prisoner i first opens box i . If it contains label j , the prisoner next opens box j , and continues following the labels in this way for at most 50 boxes.

This procedure follows the cycle of π containing i . Therefore every prisoner finds their own label if and only if every cycle of π has length at most 50.

For $51 \leq k \leq 100$, the probability that a uniformly random permutation contains a k -cycle is $1/k$. Moreover, two cycles of lengths greater than 50 cannot occur simultaneously. Hence the probability that some cycle is too long is

$$\sum_{k=51}^{100} \frac{1}{k}.$$

The probability that all prisoners survive is therefore

$$1 - \sum_{k=51}^{100} \frac{1}{k} = 1 - (H_{100} - H_{50}) \approx 0.3118278207.$$

Thus the cycle-following strategy saves all the prisoners with probability approximately

$$\boxed{31.18\%},$$

which is greater than 30%.

□

Problem 12. (St. Petersburg Paradox). A fair coin is tossed until heads appears for the first time. If the first head appears on the n th toss, the player receives

$$2^n$$

units of money. Compute the expected payoff. If decisions are based only on expected monetary value, determine how large an entry fee it would be rational to pay, and explain why the conclusion is regarded as paradoxical.

Solution. The probability that the first head appears on the n th toss is

$$\left(\frac{1}{2}\right)^n.$$

Therefore the expected payoff is

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} 2^n \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} 1 = \boxed{\infty}.$$

Under the rule of maximizing expected monetary value alone, subtracting any finite entry fee still leaves an infinite expected net payoff. Thus there is no finite upper bound on the fee that this rule recommends.

The conclusion conflicts sharply with ordinary judgment. For example, the payoff is at most 2 with probability $1/2$ and at most 4 with probability $3/4$, so most outcomes are small. The infinite expectation is driven by extremely rare but extremely large prizes. This mismatch shows that expected monetary value alone is not always an adequate decision criterion; finite resources, risk, and diminishing marginal utility of money can all change a rational valuation.

□

Problem 13. (Bachet's Weights Problem). Using a two-pan balance and weights of positive integer numbers of pounds, determine the minimum number of weights needed to weigh every integer load from 1 to 40 pounds. A weight may be placed on either pan. Determine the weights as well.

Solution. With m weights, each weight has three possible roles: on the load pan, unused, or on the opposite pan. Hence at most 3^m signed sums can be produced. These include 0 and occur in opposite pairs, so at most

$$\frac{3^m - 1}{2}$$

positive loads can be distinguished. To cover $1, 2, \dots, 40$, we need

$$\frac{3^m - 1}{2} \geq 40,$$

and therefore $m \geq 4$.

Four weights suffice: take

$$\boxed{1, 3, 9, 27}.$$

Every integer from -40 to 40 has a balanced ternary representation

$$N = \varepsilon_0 + 3\varepsilon_1 + 9\varepsilon_2 + 27\varepsilon_3, \quad \varepsilon_i \in \{-1, 0, 1\}.$$

A coefficient 1 places the corresponding weight opposite the load, a coefficient -1 places it with the load, and a coefficient 0 leaves it unused. Thus all loads from 1 through 40 can be weighed, and four weights are minimal.

□

Problem 14. (Hex). On a finite $n \times n$ Hex board, two players alternately place stones of their own colors in empty cells. Each player wins by forming a connected path joining a designated pair of opposite sides. Prove that the first player has a winning strategy.

Solution. A completely filled Hex board cannot be a draw. Indeed, consider the component of one color attached to one of its target sides. If that component does not reach the opposite target side, its boundary across the hexagonal cells produces a connected chain of the other color between the other pair of opposite sides. Thus exactly one player has a winning connection when the board is full.

Since the game is finite and has no draws, one of the two players has a winning strategy. Suppose, for contradiction, that the second player has one. The first player places a stone in any cell and thereafter follows the supposed second-player strategy, using the symmetry that interchanges the two colors and their target sides. If the strategy ever calls for the already occupied initial cell, the first player makes any legal move instead and continues.

An extra stone of one's own color can never destroy a connection, so this stolen strategy still wins. That contradicts the assumption that the second player had a winning strategy. Hence the first player has one. The argument is nonconstructive and need not identify a winning opening move.

□

Problem 15. (Secretary Problem). There are $n \geq 2$ applicants with distinct ranks, presented in a uniformly random order. After each interview, the current applicant must be accepted or rejected permanently. Only the relative ranking of the applicants seen so far is known. Determine a strategy maximizing the probability of selecting the best applicant, and find its asymptotic success probability as $n \rightarrow \infty$.

Solution. Only an applicant who is best among those seen so far can be accepted optimally. The optimal rule has a threshold form: reject the first r applicants, then accept the first subsequent applicant who is better than all of the first r .

For a fixed r , the best applicant must occur at some position $k > r$, and then the best among the first $k - 1$ applicants must lie among the first r . Therefore the success probability is

$$P_n(r) = \sum_{k=r+1}^n \frac{1}{n} \frac{r}{k-1} = \frac{r}{n} \sum_{j=r}^{n-1} \frac{1}{j}.$$

Moreover,

$$n(P_n(r+1) - P_n(r)) = \sum_{j=r+1}^{n-1} \frac{1}{j} - 1.$$

Hence $P_n(r)$ increases until the displayed harmonic tail passes below 1 and then decreases. Thus an optimal cutoff is the integer where that crossing occurs.

Since

$$\sum_{j=r}^{n-1} \frac{1}{j} = \ln\left(\frac{n}{r}\right) + o(1),$$

the optimal cutoff satisfies

$$\frac{r}{n} \rightarrow \frac{1}{e}.$$

Substitution in $P_n(r)$ gives the limiting success probability

$$\boxed{\frac{1}{e}}.$$

□

Problem 16. (Nim, Bouton's Theorem). Several heaps contain

$$a_1, a_2, \dots, a_m$$

stones. Two players alternately choose one heap and remove any positive number of stones from it. The player taking the last stone wins. Characterize the positions from which the next player has a winning strategy.

Solution. Let

$$s = a_1 \oplus a_2 \oplus \dots \oplus a_m$$

be the bitwise XOR, or nim-sum, of the heap sizes.

Suppose first that $s = 0$. Any legal move changes one heap from a_i to a smaller value b_i . At the highest binary digit where a_i and b_i differ, the new nim-sum has a 1, so every move leads to a position with nonzero nim-sum.

Now suppose that $s \neq 0$. Choose the highest binary digit in which s has a 1. Some heap a_i also has a 1 in that position. Set

$$b_i = a_i \oplus s.$$

At the highest chosen digit, b_i has 0 while a_i has 1, so $b_i < a_i$ and this is a legal reduction. The new nim-sum is

$$s \oplus a_i \oplus b_i = 0.$$

Thus every nonzero nim-sum position can move to a zero nim-sum position, while every move from a zero nim-sum position makes it nonzero. Therefore

$$a_1 \oplus \dots \oplus a_m \neq 0 \text{ exactly for the next-player winning positions.}$$

□

Problem 17. (100 Prisoners and a Light Bulb). One hundred prisoners are isolated. Each day, one prisoner is selected independently and uniformly at random and taken to a room containing a light bulb, initially off. A prisoner in the room may inspect and change the bulb. Before the process begins, the prisoners may agree on a strategy. At any visit, a prisoner may declare that everyone has visited the room; a correct declaration frees all prisoners, while an incorrect one kills them. Find a strategy that eventually permits a correct declaration with probability 1.

Solution. Designate one prisoner as the counter. Each of the other 99 prisoners follows this rule: the first time that prisoner enters and finds the bulb off, turn it on; after having done so once, never turn it on again.

The counter never turns the bulb on. Whenever the counter enters and finds it on, the counter turns it off and increases a private count by 1. When the count reaches 99, the counter declares that everyone has visited.

Every counted signal was produced by a different non-counter, so the declaration is certainly correct. Under independent uniform random selection, every prisoner is selected infinitely often with probability 1. Consequently, each uncounted prisoner eventually encounters an off bulb and signals, and the counter eventually records every signal. Thus the correct declaration occurs with probability 1.

□

Problem 18. (Sicherman Dice). Label the six faces of each of two dice by positive integers, allowing repeated labels. Find a pair of dice, not both standard, whose sum has exactly the same probability distribution as the sum of two standard dice.

Solution. For a die with face labels $d_1, \dots, d_6$, encode its sums by the polynomial

$$P(x) = x^{d_1} + \dots + x^{d_6}.$$

The coefficient of x^k in the product of the two dice polynomials is the number of outcomes with sum k .

Take dice with labels

$$\boxed{1, 2, 2, 3, 3, 4}$$

and

$$\boxed{1, 3, 4, 5, 6, 8}.$$

Their polynomials are

$$P(x) = x + 2x^2 + 2x^3 + x^4$$

and

$$Q(x) = x + x^3 + x^4 + x^5 + x^6 + x^8.$$

Direct factorization gives

$$P(x)Q(x) = (x + x^2 + x^3 + x^4 + x^5 + x^6)^2.$$

Hence every possible sum occurs with exactly the same multiplicity as for two standard dice. $\square$

Problem 19. (Number Erasing Game). The integers

$$1, 2, \dots, 100$$

are written on a board. Repeatedly erase two numbers a, b and replace them by

$$a + b + ab.$$

Determine the final number and prove that it is independent of all choices.

Solution. The operation is designed so that

$$(a + b + ab) + 1 = (a + 1)(b + 1).$$

Thus if every current board entry x is replaced conceptually by $x + 1$, each move simply replaces two factors by their product. The product of all shifted entries is therefore invariant.

Initially this product is

$$\prod_{k=1}^{100} (k + 1) = 101!.$$

If the final entry is N , the final shifted product is $N + 1$. Hence

$$N + 1 = 101!,$$

and the last number is

$$\boxed{101! - 1}.$$

$\square$

Problem 20. (Sprouts). Start with n spots in the plane. A move joins two spots, possibly the same spot, by a non-self-intersecting curve that crosses no existing curve, places one new spot on the new curve, and keeps every spot's degree at most 3. Prove that the game cannot continue indefinitely.

Solution. Call each unused incidence available at a spot a *life*. A spot of degree d has $3 - d$ lives. Initially there are $3n$ lives.

Every move uses two lives at its endpoints. The new spot lies in the interior of the new curve and immediately has degree 2, so it creates one new life. Therefore each move decreases the total number of lives by exactly

$$2 - 1 = 1.$$

Since the number of lives is always nonnegative, there can be at most $3n$ moves. In particular, every game of Sprouts terminates. □

Problem 21. (Seven Hats Problem). Seven players are independently given red or blue hats, each color having probability $1/2$. Every player sees the other six hats but not their own. Simultaneously, each player guesses red, guesses blue, or passes. The group wins if at least one guess is correct and no guess is incorrect. Find a strategy with winning probability $7/8$.

Solution. Encode red and blue by 0 and 1, so a hat configuration is a vector in $\mathbb{F}_2^7$. Let $\mathbf{H}$ be the 3×7 matrix whose columns are the seven nonzero vectors of $\mathbb{F}_2^3$, and let

$$C = \{\mathbf{x} \in \mathbb{F}_2^7 \mid \mathbf{H}\mathbf{x} = \mathbf{0}\}$$

be the binary Hamming code. Since $\mathbf{H}$ has rank 3, the code has $2^{7-3} = 16$ words. No nonzero codeword has weight 1 or 2, because the columns of $\mathbf{H}$ are nonzero and distinct; hence the minimum distance is at least 3. The radius-1 Hamming balls around the 16 codewords are disjoint and contain

$$16(1 + 7) = 128$$

vectors in total. They therefore partition $\mathbb{F}_2^7$, so every non-codeword differs from a unique codeword in exactly one coordinate.

Player i considers the two full configurations consistent with the six visible hats. If exactly one is a codeword, player i guesses the opposite of the i th bit of that codeword. Otherwise, player i passes.

If the actual configuration is a codeword, all seven players guess the opposite of their actual hats, so the group loses. If the actual configuration is not a codeword, let the unique nearest codeword differ in coordinate i . Player i guesses correctly, while every other player passes because changing another coordinate would produce a word at distance 2 from that codeword, not another codeword. Thus the group wins on all $128 - 16 = 112$ non-codewords. The success probability is

$$\boxed{\frac{112}{128} = \frac{7}{8}}.$$

□

Problem 22. (Circular Gas Station Problem). Around a circular road are n gas stations. At station i , a car gains g_i units of fuel, and the trip to the next station costs c_i units. Suppose that

$$\sum_{i=1}^n g_i \geq \sum_{i=1}^n c_i.$$

Prove that some station permits a car with an initially empty tank to complete one full circuit without ever running out of fuel.

Solution. Put

$$d_i = g_i - c_i, \quad S_0 = 0, \quad S_k = \sum_{i=1}^k d_i.$$

Choose $j \in \{0, 1, \dots, n-1\}$ s.t. S_j is minimal, and start at station $j+1$, with indices understood cyclically.

Before wrapping around the circle, the fuel after t legs is

$$S_{j+t} - S_j \geq 0.$$

After wrapping around, the corresponding amount has the form

$$S_n + S_k - S_j.$$

Here $S_n \geq 0$ by hypothesis and $S_k \geq S_j$ by the minimality of S_j , so this quantity is also nonnegative. Hence the car completes the full circuit from station $j+1$. □

Problem 23. (2013 Oxford MAT). Charlie writes one integer from 1 to 10 on Alice's forehead and one on Bob's forehead. The two integers differ by 1. Alice and Bob can see each other's number but not their own, and all three people always tell the truth. They have the following conversation:

Alice: "I do not know my number. Will you tell me whether my number is a square?"

Charlie: "If I tell you that, then you will know your number."

Bob: "I still do not know my number."

Determine Alice's number.

Solution. Let Bob's number be b . Alice initially does not know her number, so $b \neq 1, 10$; otherwise only one adjacent integer would be allowed.

Charlie's statement means that among Alice's two candidates

$$b-1 \quad \text{and} \quad b+1,$$

exactly one is a square. The squares from 1 to 10 are

$$1, 4, 9,$$

so this condition leaves

$$b \in \{2, 3, 5, 8\}.$$

Bob sees Alice's number a and still cannot determine his own number. Under the preceding restriction, this ambiguity occurs only when

$$a = 4,$$

because then Bob's number could be either 3 or 5. For every other possible value of a , exactly one adjacent number belongs to $\{2, 3, 5, 8\}$. Hence Alice's number is

$$\boxed{4}.$$

□

Problem 24. (1998 All-Russian MO). An integer from 1 to 144 is chosen. One may repeatedly choose any subset S and ask whether the chosen integer belongs to S . A “Yes” answer costs 2 rubles, while a “No” answer costs 1 ruble. Determine the minimum cost that is sufficient in the worst case to identify the chosen integer.

Solution. Let $N(c)$ be the greatest number of possibilities that can always be distinguished with remaining worst-case cost at most c . With no useful question available,

$$N(0) = N(1) = 1.$$

For a first question, the “No” branch has remaining budget $c - 1$ and the “Yes” branch has remaining budget $c - 2$. Therefore

$$N(c) = N(c - 1) + N(c - 2).$$

Conversely, this bound is attainable by partitioning the current set of possibilities into subsets of these two maximum sizes and asking whether the chosen number lies in the second subset.

Thus

$$N(c) = F_{c+1},$$

where $F_1 = F_2 = 1$. Since

$$N(10) = F_{11} = 89 < 144$$

but

$$N(11) = F_{12} = 144,$$

the minimum guaranteed cost is

$$\boxed{11 \text{ rubles}}.$$

□

Problem 25. (1976 Waseda Entrance Examination). One card is chosen from a standard deck of 52 cards and placed in a box without being seen. The remaining 51 cards are shuffled, and three cards are drawn. All three drawn cards are diamonds. Determine the conditional probability that the card in the box is a diamond.

Solution. Regard the entire experiment as placing the 52 shuffled cards into labeled positions, with the first position designated as the box and the next three positions designated as the observed draws. After the three observed cards are known to be diamonds, the remaining 49 unobserved positions contain exactly

$$10$$

diamonds. By symmetry, each of these 49 positions is equally likely to be the box position. Therefore the desired conditional probability is

$$\boxed{\frac{10}{49}}.$$

□

Problem 26. A traditional epitaph describes the life of Diophantus as follows. He spent one sixth of his life as a child, one twelfth more before growing a beard, and one seventh more before marrying. Five years later his son was born; the son lived half as long as Diophantus, and Diophantus died four years after his son. Determine Diophantus’ age at death.

Solution. Let x be Diophantus' age at death. The epitaph gives

$$x = \frac{x}{6} + \frac{x}{12} + \frac{x}{7} + 5 + \frac{x}{2} + 4.$$

Since

$$\frac{1}{6} + \frac{1}{12} + \frac{1}{7} + \frac{1}{2} = \frac{25}{28},$$

we obtain

$$\frac{3x}{28} = 9.$$

Therefore,

$$\boxed{x = 84}.$$

□

Problem 27. In a country where every statement made by an inhabitant is false, five people A, B, C, D, E have distinct ages and say:

- (a) A : "I am the oldest."
- (b) B : "I am the youngest."
- (c) C : "I am older than D ."
- (d) D : "I am older than E ."
- (e) E : "I am younger than B ."

Who is the oldest?

Solution. Since every statement is false, C is younger than D , D is younger than E , and E is older than B . Hence

$$C < D < E, \quad B < E.$$

Also A is not the oldest. If $A > E$, then A would be older than B, C, D, E and therefore would be the oldest, contradicting A 's false statement. Thus $A < E$ as well. Consequently E is older than all four others, so

$$\boxed{E}.$$

□

Mathematical Intuition Problems

Problem 1. A geodesic triangle is drawn on a surface of constant Gaussian curvature. Predict whether its angle sum is less than, equal to, or greater than π in each case.

- (a) negative curvature;
- (b) zero curvature;
- (c) positive curvature.

Solution. By the Gauss–Bonnet relation

$$\alpha + \beta + \gamma - \pi = K \text{ area}(\triangle ABC),$$

the sign of the angle excess is the sign of K . Therefore,

- (a) less than π ,
- (b) equal to π ,
- (c) greater than π .

□

Problem 2. (1996 Putnam). Two circles have radii 1 and 3, and the distance between their centers is 10. A point X moves on the first circle and a point Y moves on the second. Determine the locus of the midpoint M of $\overline{XY}$.

Solution. Let the centers of the two circles be A and B , and let O be the midpoint of $\overline{AB}$. Write

$$\overrightarrow{AX} = \mathbf{u}, \quad \overrightarrow{BY} = \mathbf{v},$$

where

$$\|\mathbf{u}\| = 1, \quad \|\mathbf{v}\| = 3.$$

Since M is the midpoint of X and Y ,

$$\overrightarrow{OM} = \frac{\mathbf{u} + \mathbf{v}}{2}.$$

The possible lengths of $\mathbf{u} + \mathbf{v}$ range from

$$3 - 1 = 2 \quad \text{to} \quad 3 + 1 = 4.$$

Every intermediate length and every direction occur by varying the angle between $\mathbf{u}$ and $\mathbf{v}$ and rotating both vectors. Therefore the locus is the closed annulus

$$1 \leq OM \leq 2.$$

□

Problem 3. (1996 Putnam). Two squares have total area 1. Their sides, and the sides of a containing rectangle, must remain parallel, and the interiors of the two squares may not overlap. Determine the least number A s.t. every such pair of squares can be packed into a rectangle of area at most A .

Solution. Let the side lengths be $x \geq y > 0$, so

$$x^2 + y^2 = 1.$$

Since two axis-parallel squares with disjoint interiors must be separated horizontally or vertically, any containing rectangle has one side at least $x + y$ and the other at least x . This bound is attained by placing the squares side by side. Thus the least area required for this pair is

$$x(x + y).$$

Put $t = y/x$, where $0 < t \leq 1$. Then

$$x^2 = \frac{1}{1 + t^2},$$

and the required area is

$$\frac{1 + t}{1 + t^2}.$$

Differentiation shows that this expression is maximized at

$$t = \sqrt{2} - 1.$$

Its maximum value is

$$\boxed{\frac{1 + \sqrt{2}}{2}}.$$

□

Problem 4. (1994 CSAT). A cube is intersected by a plane. Among the following shapes, determine how many can occur as the cross-section:

- (1) a triangle;
- (2) a nonsquare rectangle;
- (3) a nonsquare rhombus;
- (4) a pentagon;
- (5) a hexagon.

Solution. Work with the cube

$$0 \leq x, y, z \leq 1.$$

Each of the five shapes occurs.

The plane

$$x + y + z = \frac{1}{2}$$

produces a triangle. The plane

$$z = \frac{x}{2}$$

produces a rectangle whose side lengths are 1 and $\sqrt{5}/2$, so it is not a square.

The plane

$$4z = x + y + 1$$

produces a parallelogram with adjacent side vectors

$$\left(1, 0, \frac{1}{4}\right) \quad \text{and} \quad \left(0, 1, \frac{1}{4}\right).$$

These vectors have equal lengths but nonzero dot product, so the cross-section is a nonsquare rhombus.

The plane

$$x + y + 2z = \frac{11}{10}$$

meets five edges of the cube, at

$$\left(\frac{1}{10}, 1, 0\right), \left(1, \frac{1}{10}, 0\right), \left(1, 0, \frac{1}{20}\right), \left(0, 0, \frac{11}{20}\right), \left(0, 1, \frac{1}{20}\right),$$

and therefore produces a pentagon. Finally,

$$x + y + z = \frac{3}{2}$$

produces a hexagon. Thus all five listed shapes are possible, and the required number is

$$\boxed{5}.$$

□